

Admissible Construction and the Non-Instantiability of Gödel Architectures

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Abstract

This paper does not contest Gödel’s incompleteness theorems in their standard metamathematical setting. It proves instead a narrower structural theorem: no Gödel-style *same-domain* incompleteness architecture is instantiable inside a fixed admissibility-governed construction regime whose domain-defining structure already includes admissibility, standing-bearing reuse, identity-preserving transformation, unique admissible interior, coherent legal trajectory structure, and irrecoverable failure. The downstream interface, quotient, and trajectory layers are received as consequences of that complete original substrate, with their licensed uses and warrants inherited through faithful representation. The decisive bridge theorem shows that constructional behavior on the fixed domain is exhausted by standing classification and coherent legal trajectory structure. Accordingly, any same-domain Gödel-style threat must survive into one of those loci. Once that bridge is in force, construction-derived classifiers collapse to hidden gate work, same-domain diagonalization fails, semantic / infinitary / meta-foundational / richer-same-domain rescue routes fail or become explicit scope change, and no same-domain incompleteness-objection route remains. The target is not bare trace classification, construction inscription, or formal calculus as such. Formal systems matter here only insofar as they are used as construction systems inside the fixed domain.

Scope clarification. This paper is not a general thesis about construction systems or formal calculi. It analyzes admissible construction on a fixed domain. A formal system enters the argument only when it is used as a construction system whose outputs are asked to carry same-domain admissibility, standing, or licensed continuation force. In addition, the admissibility boundary and standing classification are treated throughout as *domain-defining structure*: altering them changes the domain rather than preserving it under a harmless redescription.

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1 Introduction

This paper begins from admissible construction on a fixed domain, not from formal calculi or reasoning procedures as such. The domain identity at issue already includes its admissibility boundary, standing classification, identity-preserving transformation discipline, and lawful same-domain continuation structure. A formal system matters here only insofar as it is used as a construction system inside that fixed domain. The present question is therefore not whether Gödel-style metamathematics exists in its ordinary setting, but whether a Gödel-style construction can be realized as a same-domain closure threat inside an already admissibility-governed construction regime.

Classical incompleteness remains intact where it ordinarily lives: in standard metamathematical settings with arithmetized syntax, internal classifiers, and diagonal construction. The question here is narrower. Suppose one already works inside a fixed same-domain regime of admissible construction: standing and admissibility are bivalent, the admissible interior is unique, identity-preserving transformation is tensor-preserving, coherent trajectories are pointwise standing-preserving legal paths, no deferred coherence is possible, terminal standing fully certifies coherence, and no same-scope repair, generator, or independent authorizer survives within the regime. Can a Gödel-style threat

architecture still be instantiated from within that same domain as a governance-relevant closure threat?

This paper answers no.

The proof strategy is eliminative rather than revisionary. It does not deny the classical incompleteness theorems. It asks instead what a same-domain incompleteness objection would have to do in order to matter *here*. Such an objection would have to change constructional behavior on the original fixed domain: what may be admitted, rejected, reused, fixed, or licensed as same-domain continuation. The central *Bridge Theorem (Constructional Form)* therefore shows that constructional behavior is exhausted by two and only two loci: standing classification of evaluated acts, and coherent legal trajectory structure of same-domain histories. The trajectory layer is decisive here. Once coherence is pointwise standing-preservation, once incoherent prefixes remain incoherent under admissible continuation, and once terminal standing fully certifies trajectory coherence, there is no third hidden path-locus in which a same-domain closure threat can matter. This is also why the target is not bare trace classification. A syntax, coding, or construction inscription matters in the present paper only if its same-domain re-entry changes standing classification or coherent legal trajectory structure. If it does neither, it is not a closure threat but bookkeeping about a closure threat.

The body then applies that bridge mechanically. Governance-relevant construction-derived classifiers become hidden gate work. Same-domain negative fixed-point routes fail because no closure-relevant distinction survives their re-entry. Semantic truth, reflection, infinitary rule-power, meta-foundational re-anchoring, and richer same-domain extension do not rescue the architecture: they either collapse to bookkeeping, import illicit same-domain authority, or become explicit scope change. The terminal theorem therefore states not that Gödel is false, but that no Gödel-style *same-domain constructional threat* is instantiable inside the unique admissible interior once the fixed constructional kernel is in force.

A full technical appendix package now supports the primary manuscript directly, while the fully armored audit edition is retained separately as an audit companion rather than being allowed to dominate the visible body. Formalization is treated in the same way: the prose theorem chain is primary, and detailed theorem-ledger provenance is relegated to the audit companion.

2 Minimal roadmap and kernel-relative posture

This manuscript is *kernel-relative*. It does not re-derive the kernel from below. Appendix A supplies the full technical kernel floor that distinguishes bare trace checking from constructional governance, excludes weaker verdict regimes, fixes kernel attachment at constructional governance, and records local kernel closure. These appendices are not optional background. They expose the minimal structure required for admissible construction itself, and therefore the minimal constructional substrate required for the theorem proved in the body. Relative to that floor, the interface, quotient, and trajectory layers are treated here as forced consequences of the same kernel rather than as parallel foundations.

The dependency order is strict:

1. full technical Appendix A floor;
2. forced downstream interface, quotient, and trajectory consequences;
3. constructional-behavior bridge;

4. code-role exhaustion;
5. diagonal-route failure and rescue-route exhaustion;
6. terminal same-domain non-instantiability theorem.

This ordering matters because the trajectory layer defeats the hardest hostile attack. The attack says that a same-domain Gödel threat might matter only at hidden intermediate proof paths or proof-theoretic strength levels without surfacing in standing classification or coherent trajectory behavior. Once coherence is fixed as pointwise standing-preservation along legal same-scope trajectories, once no-deferred-coherence is in force, once terminal standing certifies coherence, and once all coherence-relevant distinctions factor through tensor content rather than skin, that hostile option is gone.

Detailed formalization ledgers, long hostile-referee crosswalks, and the hyper-armored dependency reservoir are intentionally removed from the primary manuscript package and retained in the audit companion. The main paper therefore keeps only what is needed for the body path and the full technical appendices A–E.

The visible body path is therefore exact and finite. The manuscript now asks the reader to track only the following eight results in order: constructional-behavior exhaustion, closure-relevance transmission, construction-derived-classifier role exhaustion, the necessary package for a same-domain Gödel threat, no same-domain diagonalization, rescue-route exhaustion, exhaustion of same-domain incompleteness-objection routes, and the terminal non-instantiability theorem. Everything else has either been placed in the full technical appendices A–E or relegated to the audit companion.

Current foundational receipt. The original dependence order is retained. Kernel necessity is the starting floor. The current Kernel, Structure, AMetric, ATS, and Unus results make explicit the licensed joint interface, complete standing content, and unique interior already used here [1, 2, 3, 4, 5]. The historical trajectory source remains the source of the received lineage theorem [6]; its present support is identified in Appendix D. The recent Metricity and Fixation applications are not additional premises of the present result.

3 What a same-domain Gödel threat would require

3.1 Target regime

Definition 3.1 (Fixed domain and same-domain threat). A *fixed domain* is a regime whose identity already includes its admissibility boundary, standing-bearing reuse conditions, lawful same-scope transitions, and closure-sensitive continuation structure. These are domain-defining structure, not optional attributes layered onto an otherwise independent domain. A *same-domain threat* is any distinction, construction, or architecture that, without explicit scope change, changes what may be admitted, rejected, reused, committed, or licensed as same-domain continuation on that fixed domain.

Definition 3.2 (Constructional behavior). The *constructional behavior* of the fixed domain is the totality of all same-domain admissions, rejections, standing-bearing reuses, fixations, and legal same-domain continuations supported by that domain.

Definition 3.3 (Construction-derived classifier layer, re-entry, and governance relevance). Let A be a class of syntax-bearing acts on the fixed domain and let C be a class of code objects. A *same-domain construction-derived classifier layer* is a coding map $\ulcorner \cdot \urcorner : A \rightarrow C$. For a predicate $P \subseteq C$, its *re-entry* to the original acts is

$$\hat{P}(a) :\Longleftrightarrow P(\ulcorner a \urcorner).$$

The predicate P is *governance-relevant* if its re-entry changes constructional behavior on the fixed domain.

Complete original use and faithful receipt. The evaluated act retains its original bearer, context, specified question, and required support and history. A non-degenerate redescription preserves the complete original governing use, including application domains, joint conditions, and warranted outcomes. These duties are inherited through faithful ATS bookkeeping; representing them again does not impose fresh qualification. A view can display less while retaining access to the complete original content. A projection that actually loses a required governing answer is not a faithful rival in this class.

This application has a direct semantic proof. Equal retained original bearer and complete-query data identify the same independently fixed result incidence. Each original licensed reader supplies its own warranted complete outcome at that incidence. Determinacy therefore gives equality of results and of definedness, collectively for all original questions; see Kernel, Theorems 6.2–6.3 and 8.7, and Structure, Theorem 5.2 [1, 2]. The proof uses the original identity and warrants, not a new field manufactured from the desired answers. ATS, Theorems 3.3, 3.5, and 7.5 and Corollary 7.6, records the resulting complete content and joint-use transport [4].

Proposition 3.4 (Recorded trajectory-kernel consequences). *The present paper works over a fixed kernel in which the following already hold on the same fixed domain:*

- (K1) *coherence of a legal same-domain trajectory is pointwise standing-preservation along that trajectory;*
- (K2) *no deferred coherence: an incoherent prefix remains incoherent under every admissible continuation;*
- (K3) *endpoint certification: terminal standing fully certifies coherence of a legal same-domain trajectory;*
- (K4) *tensor determination and skin neutrality: all coherence-relevant distinctions factor through standing quotient and tensor content, while skin carries no constructional authority; and*
- (K5) *standing conservation and irrecoverability: there is no same-scope creation, amplification, repair, or recovery of standing inside the regime.*

Remark 3.5 (Why theorem 3.4 matters). These are the exact trajectory-layer results that defeat the hardest hostile attack. The attack says that a same-domain Gödel threat might hide in intermediate path structure without surfacing in standing classification or terminal trajectory behavior. Theorem 3.4 rules that out: pointwise standing-preservation, no-deferred-coherence, endpoint certification, and tensor determination leave no independent hidden path-locus with governance force.

Theorem 3.6 (Bridge Theorem (Constructional Form): constructional-behavior exhaustion). *Constructional behavior on the fixed domain is exhausted by two and only two loci:*

- (i) *the standing classification of evaluated acts; and*
- (ii) *the coherent legal trajectory structure of same-domain histories, with coherence equivalently given, on those already legal retained lineages, by pointwise standing-preservation and terminal standing behavior.*

If two same-domain descriptions agree on those loci, then they agree on what may be admitted, rejected, reused, fixed, or licensed as same-domain continuation.

Proof. By theorem 3.4, any same-domain act that matters to admission, rejection, or reuse already appears in the standing classification of evaluated acts. Likewise, any same-domain lineage that matters to closure-sensitive continuation is governed by the coherent-construction layer: coherence is pointwise standing-preservation on legal same-domain trajectories, no deferred coherence exists, and terminal standing certifies coherence. Therefore any governance-relevant difference in constructional behavior must appear either as a difference in standing classification or as a difference in coherent legal trajectory structure. Since skin carries no constructional authority and no second same-domain standing-bearing equivalence survives, there is no third governance-bearing locus left inside the regime. $\square$

The trajectory locus includes the original domain of legal paths, their applicability and joint warrant conditions, and the retained history. Endpoint certification compresses the coherence answer on an already legal retained lineage; it does not erase those other questions or establish legality from arbitrary statewise agreement. Appendix D makes the inherited state and transition identification explicit.

Lemma 3.7 (Closure-relevance transmission). *If a same-domain distinction changes constructional behavior, then it changes either the standing classification of some evaluated act or the coherent legal trajectory / terminal-standing behavior of some legal same-domain trajectory.*

Proof. This is the contrapositive reading of theorem 3.6. If a distinction changed neither standing classification nor coherent legal trajectory structure, then it would change nothing about what may be admitted, rejected, reused, or legally continued on the fixed domain. It would therefore be closure-inert. $\square$

Theorem 3.8 (Necessary package for a same-domain Gödel threat). *If a Gödel-style phenomenon is to threaten same-domain closure from within the original domain, it must realize all of the following together:*

- (i) *a faithful same-domain coding layer;*
- (ii) *a governance-relevant construction-derived classifier or re-entry condition;*
- (iii) *a negative fixed-point or diagonal route capable of producing a same-domain closure-sensitive distinction; and*
- (iv) *survival of that distinction into standing classification or coherent legal trajectory structure on the original domain.*

Proof. A same-domain incompleteness objection matters only if it changes constructional behavior. By theorem 3.7, that requires survival into standing classification or coherent legal trajectory structure. But no such change can arise unless there is a code layer, a governance-relevant

construction-derived classifier, a same-domain re-entry, and a negative fixed-point route capable of feeding the construction-derived classifier back onto original-domain content. Remove any one of these and the architecture is either external metamathematics, mere bookkeeping, or a non-threatening definitional restatement. $\square$

4 Governance-relevant construction-derived classifiers are hidden gate work

Theorem 4.1 (Construction-derived classifier role exhaustion). *Every same-domain construction-derived classifier is exhausted by the following four role descriptions:*

- (i) *inert bookkeeping;*
- (ii) *extensional gate renaming;*
- (iii) *illicit same-domain refinement of the positive or negative side of the gate; or*
- (iv) *independent same-domain authorizer.*

Only the first two survive, and neither generates a new closure threat. Gate renaming is itself closure-inert; the two surviving descriptions need not be disjoint.

Proof. Take any same-domain construction-derived classifier P . If re-entry of P changes no standing classification and no coherent legal trajectory behavior, then by definition it is inert bookkeeping. If its positive and negative extension coincide exactly with the existing standing partition, it is extensional gate renaming.

Suppose instead that P matters nontrivially. By theorem 3.7, it must change standing classification or coherent legal trajectory behavior. If it marks a nontrivial subset of what is already admitted or already rejected as specially governance-bearing, then it refines one side of the gate. But the kernel excludes second same-domain standing-bearing refinements: same-side refinements and second equivalence relations collapse. If P instead purports to authorize or block original-domain acts independently of the existing standing partition, then it functions as an independent authorizer, which the kernel also excludes. These four cases exhaust the possibilities. $\square$

Here *inert* is extensional relative to the original governing behavior, not a synonym for useless. A reader may reveal a licensed type distinction, recover a required application domain, or provide a useful certificate while changing no original answer. Such a reader is already in the first branch. It is not placed in the illicit-refinement branch merely because its predicate differs from the admission predicate. The interface used in the proof is the original base together with its status presentation; Appendix B displays the inherited factorization through that joint interface.

Corollary 4.2 (No independent governance-relevant classifier). *No construction-derived classifier survives on the fixed domain as an independent governance-relevant closure determinant.*

Proof. By theorem 4.1, only inert bookkeeping and extensional gate renaming survive. Neither yields an independent closure determinant. $\square$

Remark 4.3 (Why this is not mere syntax compression). Theorem 4.1 is not the claim that syntax is impossible or uninteresting. It is the claim that, inside the fixed domain, syntactic or coding distinctions matter to closure only by functioning as hidden gate work, extensional gate renaming,

or inert bookkeeping. A purely formal difference with no governance-relevant re-entry is therefore not a same-domain threat.

Corollary 4.4 (Bare syntax is not standing). *A same-domain coding or construction trace that does not survive re-entry into standing classification or coherent legal trajectory structure is syntax-only bookkeeping and not an incompleteness threat on the fixed domain.*

Proof. Immediate from theorem 4.1: any surviving governance-relevant construction-derived classifier must either collapse to the existing gate or illicitly attempt hidden gate work. If a coding difference survives neither standing classification nor coherent trajectory structure, it is inert bookkeeping. $\square$

Lemma 4.5 (Representational differences are either inert or construction-relevant). *A same-domain representational difference that matters to closure either changes standing classification / coherent legal trajectory structure or changes nothing governance-relevant at all.*

Proof. This is the immediate combination of theorem 3.7 with the skin-neutrality component of theorem 3.4. $\square$

Theorem 4.6 (No non-trivial admissible encoding multiplicity). *There is no admissible same-domain multiplicity of encodings that preserves all original-domain standing and coherent legal trajectory behavior while still yielding a new governance-relevant closure distinction.*

Proof. Assume such an encoding multiplicity exists. If it preserves all original-domain standing and coherent legal trajectory behavior, then by theorem 3.6 it preserves constructional behavior. Hence it yields no governance-relevant closure distinction. If it yields a governance-relevant closure distinction, then by theorem 3.7 it must change one of those loci after all. Contradiction. $\square$

5 Diagonalization and rescue-route failure

Theorem 5.1 (No same-domain diagonalization). *No admissible same-domain negative fixed-point route that is governance-relevant on the fixed domain is instantiable inside the fixed constructional kernel.*

Proof. Assume such a route exists. Then, by theorem 3.8, it requires a governance-relevant construction-derived classifier on the fixed domain. By theorem 4.1, that predicate must be inert bookkeeping, extensional gate renaming, illicit same-domain refinement, or independent authorizer.

It cannot be inert, because then there is no threat. It cannot be an illicit refinement or independent authorizer, because the kernel excludes hidden gate-work beyond the already fixed standing and trajectory structure. If it is extensional gate renaming, then the alleged fixed point introduces no new closure-relevant distinction beyond the already-fixed standing partition. It may describe an external coding artifact, but it does not reopen constructional behavior on the original domain. Hence no governance-relevant same-domain negative fixed-point route remains. $\square$

Corollary 5.2 (No same-domain closure-relevant negative fixed point over an internal classifier). *There is no same-domain act, sentence, or content-token whose governance relevance on the fixed domain is secured by a negative fixed point over a same-domain internal admissibility-like classifier.*

Route family	Why it fails inside the fixed domain	Disposition
Semantic truth / satisfaction / intended-model re-entry	preserves the complete original governing answers, or purports to change them through additional authority	illicit authority import or inert bookkeeping
Reflection / infinitary rule-power / transfinite closure	has any alleged change witnessed at an original act or trajectory question, where its original warrant and complete outcome must be preserved	captured already or illicit authority import
Meta-foundational re-anchoring	changes what counts as authoritative from outside the fixed domain	explicit scope change or illicit authority import
Richer same-domain extension	either adds no new closure-relevant distinction, or adds a second gate/equivalence, or silently changes the domain	inert, illicit refinement, or scope change

Table 1: Rescue routes after the kernel bridge.

Proof. Such an object would instantiate the route excluded by theorem 5.1. $\square$

A syntactic fixed point is not by itself a constructional threat. In particular, a content equivalence with $\neg\text{Prov}(\ulcorner g \urcorner)$ does not establish $\text{Adm}(g) = \neg\text{Adm}(g)$ or identify internal provability with the original standing question. A sound incomplete proof predicate may perform an already licensed certificate use. It cannot claim completeness for a different original question merely by reporting that no certificate was found. If an incompleteness argument refutes an advertised implementation's coverage, it exposes failure to realize that complete use; it is not excluded as a metamathematical objection.

6 Rescue-route exhaustion

Theorem 6.1 (Rescue-route exhaustion). *Every same-domain attempt to rescue a Gödel-style incompleteness objection by semantic truth, satisfaction, intended-model semantics, reflection, infinitary rule-power, transfinite closure, meta-foundational re-anchoring, or richer same-domain extension collapses into one of three outcomes: closure-inert bookkeeping, illicit same-domain authority import, or explicit scope change.*

Proof. Take the route families in table 1. A semantic rescue matters only if its re-entry changes constructional behavior on the original domain. By theorem 3.7, that change must survive into standing or coherent legal trajectory behavior. If the semantic layer does not do that, it is inert bookkeeping. If it does, it becomes fresh same-domain authority, which the kernel excludes.

The same argument applies to reflection and infinitary rule-power. If such machinery has no new closure-relevant effect, it is bookkeeping. If it does, then its alleged change has a witness in an original standing or trajectory question. An originally licensed complete use preserves that answer by its inherited warrant; a conflicting answer imports an authority not licensed by that use. Meta-foundational re-anchoring fails for the same reason, except that the imported authority is explicit. Finally, richer same-domain extension either preserves constructional behavior, in which

case it is inert; or alters it, in which case it introduces illicit same-domain refinement or becomes explicit scope change. No fourth rescue outcome remains. $\square$

A witness question is not thereby an effective finite verification procedure or a finite proof of a semantic answer. Originally licensed semantic or infinitary resources retain their actual warrants. The exclusion concerns a change to the complete original governance, not a resource's name. The route test also applies to an unlisted method: identify its original question and warranted readback, then apply the same behavior-preservation argument.

7 Terminal theorem and corollaries

Theorem 7.1 (Exhaustion of same-domain incompleteness objection routes). *Every same-domain incompleteness objection route falls into one of the following six families:*

- (i) *code-level re-entry routes;*
- (ii) *semantic or intended-model rescue routes;*
- (iii) *reflection, infinitary, or transfinite closure routes;*
- (iv) *second-equivalence or hidden-refinement routes;*
- (v) *richer same-domain extension or new-gate routes; and*
- (vi) *explicit scope-change routes.*

The first five are impossible or closure-inert inside the fixed constructional kernel; the sixth is not same-domain.

Proof. Code-level re-entry routes are exhausted by theorem 4.1. Any route that genuinely threatens closure must also satisfy the package in theorem 3.8; theorem 5.1 then closes the negative fixed-point branch. Semantic, reflection, infinitary, and meta-foundational rescue routes are exhausted by theorem 6.1. Any surviving second-equivalence or richer same-domain extension route would reintroduce hidden gate work or a new same-domain authority source, again excluded by theorems 4.1 and 6.1. What remains is explicit scope change, which is not a same-domain threat by definition. $\square$

Theorem 7.2 (Terminal non-instantiability theorem). *No Gödel-style same-domain threat architecture is instantiable in the unique admissible interior of the fixed constructional regime.*

Proof. Suppose such an architecture were instantiable. Then it would belong to one of the six route families of theorem 7.1. The first five are impossible or closure-inert in the fixed kernel. The sixth is explicit scope change and therefore not same-domain. This contradiction shows that no same-domain Gödel threat architecture is instantiable. $\square$

Corollary 7.3 (No same-domain “true but unprovable” split with governance force). *There is no same-domain split between a standing-bearing content and its internal provability-like classification that survives as a governance-relevant closure threat on the fixed domain.*

Proof. Any such split would require a same-domain Gödel threat architecture and is excluded by theorem 7.2. $\square$

Corollary 7.4 (Kernel-relative reading). *Theorem 7.2 is a kernel-relative non-instantiability result. It leaves classical incompleteness untouched while eliminating same-domain threat architectures inside the fixed constructional regime.*

The quantified class is the non-degenerate, faithful realization of the same complete construction. A selected view retaining complete original access remains a presentation inside the one licensed interior. A purported replacement that loses required governance is not a faithful second interior. Redundant raw labels likewise do not supply additional governing content. These are applications of the original scope conditions, not extra exclusion tests.

The true-but-unprovable corollary concerns a split that survives as a change to that original governance. It does not identify every truth question with every proof predicate or deny an ordinary formal incompleteness result. Those original questions, their answer types, and their warrants must be kept fixed through re-entry. Missing coverage cannot be repaired by substituting a report of unavailable proof for a different required conclusion.

8 Limits and non-claims

The paper does not claim that Gödel’s incompleteness theorems are false. It does not claim that every formal calculus, every construction practice, or every metamathematical coding scheme satisfies the target regime. It does not deny external metamathematics, explicit scope change, or undecidability phenomena that live outside same-domain constructional governance. It does not show that no broader completion project is possible after enlarging scope, changing the admissibility regime, or importing new standing-bearing structure. It also does not treat bare derivation-trace systems, coding layers, or external proof checkers as automatically belonging to the target. They enter the argument only if their same-domain re-entry can alter constructional behavior on the fixed domain.

Its claim is narrower and stronger inside that narrowness. Once the fixed domain is already forced into the fixed constructional kernel, all same-domain governance-relevant closure threats must survive into standing classification or coherent legal trajectory structure. The trajectory layer shows that no hidden path-level third locus remains. The code-role theorem shows that governance-relevant coding is hidden gate work. The route-exhaustion theorems show that no same-domain Gödel threat architecture survives those constraints.

Provisional practice and repaired proofs. The framework does not deny exploratory mathematics, seminar sketches, blackboard heuristics, unpublished drafts, or provisional codings. Before they are taken as standing-bearing same-domain acts, they are provisional practice or skin-level bookkeeping. Once a sketch, draft proof, or coding is entered as a standing-bearing same-domain act, however, the kernel applies immediately. A later corrected proof is therefore a fresh act, or an explicit reset of scope or target, rather than a same-act repair inside the retained lineage. This is exactly the sense in which the theory classifies repaired proofs without forbidding mathematical practice.

Formalization status. The prose theorem chain is primary. The frozen April audit companion records the historical Lean4 consequence-layer development and its theorem ledger. It is retained as

provenance, not as a machine verification of this reference update or of its added receipt arguments. No new formalization claim is made here.

A Kernel closure: necessity floor, non-derivability, and minimality

This appendix reconstructs the necessity floor needed by the main theorem. Its role is not merely to record a preferred starting vocabulary. It proves that, for the target phenomenon of construction-governed determinate construction under presentation-stable admissibility-invariant use, admissibility, standing, reference, and irreversibility are already forced, mutually closed, and not faithfully derivable from below within the same meaningful regime. The attachment point is Level 2 constructional governance rather than the stronger Level 3 closure-stable certification layer. Every later appendix and every theorem in the main body works downstream of that floor.

For hostile reading, Appendix A may be traversed in the following order: first the displayed-package layer and the claim that packages do not yet determine uptake-space; second the proof that uptake possibilities are still not constructional roles; third the realization criterion that fixes constructional governance only under package-stability, anti-drift, anti-repair, and downstream-sensitivity; fourth the concrete failure-family and weaker-uptake exclusion theorems; fifth the attachment of the kernel at Level 2 constructional governance; and finally the separation between Level 2 constructional governance and Level 3 closure-stable certification. This roadmap is expository only; it adds no new assumptions.

The argumentative posture of Appendix A is eliminative rather than reductive or stipulative. It does not claim a substrate more primitive than displayed-package use and then redescribe it in preferred vocabulary. Instead it proves that bare trace-classification, displayed packages, displayed uptake possibilities, and structurally weaker uptake regimes are all insufficient, and that what survives those exclusions is constructional governance in the precise sense fixed below. The force of the appendix is therefore in ruling out weaker verdict-regimes, not in stipulating a preferred description of ordinary proof practice or reducing governance to a thinner syntactic surrogate.

Definition A.1 (Meaningful regime and non-degenerate reasoning). A regime is *meaningful* for the present target if its outputs are assessable as construction-governed determinate construction acts under admissibility-invariant use rather than as bare symbol traces, encodings, or bookkeeping records. A construction act is *non-degenerate* if the following hold together:

- (i) **Reference:** claims refer determinately rather than equivocally or vacuously;
- (ii) **Standing:** claims are licit claim-making acts rather than mere inscriptions; and
- (iii) **Irreversibility:** invalid claim-acts cannot be repaired post hoc into valid ones while remaining the same act.

Admissibility is not a fifth independent ingredient beside these three; it names the governance boundary under which they are jointly enforceable on the same construction act.

Definition A.2 (Governance-valid derivation). A derivation is *governance-valid* if its steps are themselves admissible construction acts: each step preserves determinate reference, carries standing, and is governed by irreversibility with respect to invalidity.

Definition A.3 (Lawful equivalence and same meaningful regime). Two governance conditions P and Q are *equivalent under identity-preserving transformation*, written $P \equiv_{\text{law}} Q$, if replacing P

by Q leaves the class of governance-valid derivations unchanged. Two presentations, construction systems, or meta-systems count as occurring in the *same meaningful regime* only when translation between them preserves governed target, standing-bearing act-status, the admissibility boundary between licensed continuation and prohibited repair, and the class of governance-valid derivations up to lawful equivalence.

Definition A.4 (Derivable from below and the kernel). A governance condition P is *derivable from below* if there exists a governance-valid derivation of P whose validity does not already presuppose P , any equivalent under identity-preserving transformation condition, or any condition whose own governance-validity already requires P . The *kernel* of non-degenerate reasoning is the set

$$K := \{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\},$$

where Adm abbreviates admissibility, Stand standing, Ref reference, and Irr irreversibility.

Definition A.5 (Formal calculus, derivation-check, and claim-bridge). A *formal calculus* is a presentation-level object consisting of strings, formation rules, axiom schemata or primitive judgments, and step rules generating finite derivation-traces. A *derivation-check* is the finite verification problem “does this trace conform to the calculus?” and yields only presentation-relative rule-conformity. A *claim-bridge* is the constructional-entitlement role by which a calculus-internal verdict is used to license proposition-bearing assertion, rejection, critique, provisional reuse, or downstream constructional extension. It is not a mandatory second event after syntax; it may be realized directly by a licensed syntactic verdict. But whether a verdict genuinely realizes that role is not settled by syntax alone: it is fixed only under the invariance constraints proved below.

Definition A.6 (Three assessment levels: trace, governance, certification). A derivation-trace is *bare* when it is assessed only for presentation-relative rule-conformity. It is an *construction-governed construction act* when, through a claim-bridge, it is used as a licit claim-bearing step whose result is treated as proposition-bearing content fit for assertion, rejection, critique, provisional reuse, or downstream constructional extension. It is *closure-stable* when, in addition, its constructional status is fixed at act-time and remains stable under identity-preserving transformation and retained-lineage continuation. Thus trace-assessment, constructional governance, and closure-stable certification are distinct levels rather than a single undifferentiated notion of reasoning.

Remark A.7 (Where the kernel first attaches). Level 1 bare trace-assessment does not yet carry the kernel. The kernel first attaches at Level 2 constructional governance, where traces are used as proposition-bearing inferential licenses under admissibility-invariant use. Level 3 then adds closure-stable certification and retained-lineage hardening without relocating that attachment point.

Definition A.8 (Governance-valid act identity). The *governance-valid identity* of an act is its identity as a licensed claim-bearing contribution at act-time, not its merely historical, typographic, archival, or theorem-directed identity. Accordingly, two inscriptions may be historically grouped as “the same proof” while differing in governance-valid act identity whenever one owes its validity to a later correction, replacement, or supplementary step not available at the original act-time.

Definition A.9 (Displayed presentation package; package-preserving reformulation; drift events). A *displayed presentation package* is a quintuple

$$P = (T, B, R, S, I)$$

consisting of: a displayed conclusion-token T ; a displayed support basis B of premises, prior lines, or cited dependencies; a displayed rule/dependency specification R recording the route or rule

applications explicitly displayed in the presentation; a scope tag S fixing the presentation-regime within which the package is displayed; and a presentation-identity marker I individuating this displayed package as this package rather than another. The package is presentation-level only: it records what is explicitly displayed, not yet whether the package is licensed for inferential uptake.

A *package-preserving reformulation* is a reformulation that alters only notation, variable naming, layout, ordering of purely expository material, punctuation, line-label style, or other skin-level display features while leaving the displayed package $P = (T, B, R, S, I)$ fixed.

A *target drift* is any untracked change in T ; a *support drift* is any untracked change in B or R ; a *scope drift* is any untracked change in S ; an *identity drift* is any untracked change in I ; and a *verdict drift* is any change in the licensed/unlicensed verdict attached to a fixed displayed package under package-preserving reformulation.

A *downstream uptake profile* for a displayed package P is a publicly typed downstream treatment of P , such as citing P as available support, marking P as unavailable support, opening P for local critique or decomposition, or extending a derivation under the assumption that P is available. A verdict mechanism is *trajectory-sensitive* on P if a difference between positive and negative verdict on P changes at least one downstream uptake profile available for that same displayed package.

A redescription of candidate constructional uses is *lawful* when, relative to a fixed displayed presentation package P , package-preserving reformulation leaves fixed the displayed package itself and does not induce target drift, support drift, scope drift, identity drift, or verdict drift. It is *role-preserving* when it does not promote mere syntax, bookkeeping, parameters, formatting, or redundancy into target-determining support, and does not demote target-determining support to them. A *silent redescription* is any untracked target, support, scope, identity, or verdict drift without explicit re-admission or identity reset.

Theorem A.10 (Syntax–Standing Separation). *Well-formed syntax is neither sufficient nor presumptive for standing. Formally, for any expression E ,*

$$\text{WellFormed}(E) \not\Rightarrow \text{Stand}(E).$$

Standing requires admissible grounding together with preservation of reference under identity-preserving transformation.

Proof. Syntax produces well-formed expressions but does not, by itself, enforce referent existence, identity, persistence, or admissibility-invariant constructional use. Standing requires that what is being claimed remain stable under identity-preserving transformation in the sense of theorem A.9. No purely syntactic property entails such invariance. Therefore syntax alone neither generates standing nor entitles one to presume it. $\square$

Lemma A.11 (Derivation-checking alone does not yet yield construction-governed reasoning). *A derivation-check on a formal calculus, by itself, determines only presentation-relative conformity to rules. It does not yet determine that the checked trace is an construction-governed construction act rather than a bare trace in the sense of theorem A.6.*

Proof. A derivation-check answers a syntactic question: whether a finite sequence is well-formed relative to a stated calculus. From that answer alone one obtains only conformity-to-rules within the presentation. One does not yet obtain, without further invariance conditions, (i) which propositions are being displayed as opposed to merely encoded, (ii) that the checked lines are licit claim-steps

rather than admissible inscriptions in a game of symbol manipulation, or (iii) that a positive verdict is entitled to govern later critique, reuse, or extension. Those stronger roles are exactly what the claim-bridge names. So derivation-checking alone is insufficient for construction-governed acts, even though it may suffice for assessment of bare derivation objects. $\square$

Theorem A.12 (Non-vacuity of syntactic exclusion). *There exist syntactic verdict mechanisms on formal calculi that fail the governance-realization test. Consequently, the exclusion of non-qualifying syntactic gates is not vacuous.*

Proof. Fix any formal calculus admitting at least one rule-conformant derivation-trace and display it as a package $P = (T, B, R, S, I)$. Define a verdict mechanism V_{fmt} by: $V_{\text{fmt}}(P)$ is positive exactly when the underlying trace is rule-conformant and the final displayed line is boldfaced. Now remove the boldface while leaving T, B, R, S, I fixed. This yields a package-preserving reformulation of the same displayed package. Ordinary rule-conformity is unchanged, but the verdict of V_{fmt} flips. So V_{fmt} is syntactic yet fails package-preserving verdict stability. Therefore not every syntactic gate qualifies. $\square$

Theorem A.13 (Syntactic conformity is weaker than governance realization). *Package-preserving agreement in rule-conformity does not by itself entail constructional governance. There exist verdict mechanisms and package-preserving reformulations on which ordinary derivability is unchanged while licensed/unlicensed status differs.*

Proof. Immediate from theorem A.12. In the construction there, ordinary rule-conformity remains fixed across a package-preserving reformulation, yet the verdict changes. So syntactic conformity under a calculus is strictly weaker than governance realization. $\square$

Theorem A.14 (Inferential governance requires downstream sensitivity). *A verdict mechanism can affect constructional governance for a displayed package only if a difference between positive and negative verdict on that package changes at least one downstream uptake profile available for that same package. If verdict differences leave all downstream uptake profiles unchanged, the verdict remains bookkeeping rather than governance.*

Proof. Inferential governance concerns whether a displayed package may actually be taken up in later mathematical use. If positive and negative verdicts make no difference to any downstream uptake profile for the same package, then the verdict governs nothing beyond the trace-classification already supplied by the presentation. In that case the verdict is bookkeeping only. So downstream sensitivity is necessary for constructional governance. $\square$

Lemma A.15 (Formal-calculus uptake is bridge-directed). *Whenever a formal calculus is actually used to license conclusions, reject alternatives, certify proofs, support critique, provisionally deploy results, or govern downstream mathematical reuse, that use is no longer mere trace-classification. It is bridge-directed constructional governance in the sense of theorem A.5. Whether that bridge is genuinely realized is then settled by the realization criterion below.*

Proof. To use a calculus to license conclusions is to move from an internal rule-conformity verdict to a same-regime entitlement: the result is now taken as available for assertion, rejection, critique, provisional reuse, or further proof. That inferential-entitlement function is precisely the claim-bridge role. Without it, the calculus remains an object of study whose traces may be classified as derivable or underivable while no proposition-bearing inferential entitlement follows. Hence any live

constructional use of a calculus is bridge-directed even before one has shown that the bridge-role is stably realized. $\square$

Theorem A.16 (Displayed packages do not determine uptake-space). *A displayed package $P = (T, B, R, S, I)$ does not by itself determine any unique space of downstream uptake possibilities. Any assignment of possible downstream treatments to P is a further structural assignment beyond the mere display of the package itself.*

Proof. The displayed package fixes only what is explicitly shown: conclusion-token, support basis, displayed rule/dependency specification, scope tag, and presentation identity. Nothing in that display alone determines which later uses are to count as available, unavailable, defeasibly available, critique-open, or extension-licensing. Distinct uptake-spaces may be paired with the same displayed package without altering the package itself. Therefore uptake-space is not fixed by package-display alone but requires an additional structural assignment. $\square$

Theorem A.17 (Displayed uptake possibilities are not yet governance roles). *Let $P = (T, B, R, S, I)$ be a displayed package together with a publicly displayed list of possible downstream uptake profiles for P . By theorem A.16, assigning that list already adds structure beyond the package itself. Still, the mere listing of those possibilities is descriptive bookkeeping about how P might later be used. It is not yet constructional governance, because no verdict mechanism has yet fixed which of those possibilities are licit or unlicensed under package-stability and anti-drift constraints.*

Proof. A displayed list of possible uptake profiles for P records only candidate downstream treatments of the package. By theorem A.16, this is already an added structural assignment rather than something fixed by the package itself. But even with that assignment in place, one has not yet determined which uptake possibilities are licensed, which are blocked, or whether the resulting classification is stable under package-preserving reformulation and anti-drift constraints. So the listing of uptake possibilities is still presentation-level bookkeeping rather than constructional governance. $\square$

Theorem A.18 (Presentation-Package Stability Governance Realization Theorem). *Let V be a verdict mechanism, including a syntactic derivation-check, associated with a formal calculus Σ . Let $P = (T, B, R, S, I)$ be a displayed package of Σ together with publicly displayed downstream uptake possibilities for P . By theorem A.17, the mere display of those possibilities is not yet constructional governance. Then V counts as governance-bearing for P if and only if:*

- (i) *every package-preserving reformulation of P leaves fixed the licensed/unlicensed verdict attached to P ;*
- (ii) *any target drift, support drift, scope drift, or identity drift requires explicit re-admission or identity reset before the old verdict may be reused for the resulting package;*
- (iii) *V is trajectory-sensitive on P in the sense of theorem A.9; and*
- (iv) *no use of V permits role-shift or silent redescription to repair, replace, or reinterpret the displayed package while the result is still being treated as that same package.*

If any one of these conditions fails, V remains a presentation-relative classifier of traces or drift-tolerant bookkeeping and does not suffice for constructional governance.

Proof. By theorems A.10, A.13 and A.17, syntactic well-formedness, ordinary rule-conformity, and the bare public listing of uptake possibilities are still not enough for constructional governance. A displayed package $P = (T, B, R, S, I)$ remains a presentation-level object until a verdict mechanism stably governs which uptake possibilities are licit for that very package. If condition (i) fails, verdict drift occurs under package-preserving reformulation even though the same displayed package remains fixed; the verdict is therefore presentation bookkeeping rather than governance. If condition (ii) fails, target, support, scope, or identity drift may be treated as though the same package were still in play, so the verdict does not fix a stable object of uptake. By theorem A.14, condition (iii) is necessary: if positive and negative verdict make no difference to any downstream uptake possibility for the same package, then the verdict governs nothing. If condition (iv) fails, then role-shift or silent redescription can repair or reinterpret the package while it is still being treated as that same package, contrary to theorem A.9. Conversely, if (i)–(iv) all hold, then the verdict on P does more than accompany a public display of uptake possibilities: it stably governs which uptake possibilities are licit for that same package, and any genuine package change requires re-admission or reset rather than reinterpetive drift. In that case V counts as governance-bearing for that use. $\square$

Theorem A.19 (Failure of governance realization under package drift). *Let V be a verdict mechanism on displayed packages. If, for some displayed package P , either*

- (i) *a package-preserving reformulation of P changes the licensed/unlicensed verdict,*
- (ii) *target, support, scope, or identity drift is permitted without explicit re-admission or identity reset,*
- (iii) *verdict differences on P leave all downstream uptake profiles unchanged, or*
- (iv) *role-shift or silent redescription is permitted to repair, replace, or reinterpret P while it is still treated as that same package,*

then V fails to realize constructional governance for that use.

Proof. Clause (i) is verdict drift under package-preserving reformulation, so the verdict is presentation bookkeeping rather than governance. Clause (ii) allows package drift to be treated as though the same package were still in play, so the verdict no longer fixes a stable object of uptake. By theorem A.14, clause (iii) leaves the verdict without governance force because positive and negative verdicts govern nothing downstream. Clause (iv) permits precisely the reinterpetive repair and silent-redescription routes excluded by theorem A.9. In each case at least one constitutive condition of theorem A.18 fails. Therefore V does not realize constructional governance for that use. $\square$

Theorem A.20 (Concrete failure family for syntactic verdict mechanisms). *There are mathematically recognizable and explicitly definable families of syntactic verdict mechanisms that fail constructional governance realization. In particular:*

- (a) *for each $k \in \mathbb{N}$, let $V_{\text{fmt}}^{(k)}$ return a positive verdict exactly when the underlying trace is rule-conformant and the k -th displayed line is boldfaced; then $V_{\text{fmt}}^{(k)}$ fails constructional governance realization;*
- (b) *for each nontrivial permutation π of the displayed support basis, let V_{sup}^π return a positive verdict exactly when the underlying trace is rule-conformant and the displayed support basis occurs in the privileged order fixed by π ; then V_{sup}^π fails constructional governance realization;*

- (c) any verdict mechanism whose output depends on context-sensitive retargeting conventions fails constructional governance realization;
- (d) any verdict mechanism permitting support-basis substitutions not marked by re-admission fails constructional governance realization; and
- (e) any verdict mechanism whose positive/negative output leaves all downstream uptake possibilities unchanged fails constructional governance realization.

Proof. For (a), fix a displayed package $P = (T, B, R, S, I)$ whose underlying trace is rule-conformant and whose k -th displayed line is boldfaced. Remove the boldface while leaving T, B, R, S, I fixed. This is a package-preserving reformulation of the same package. Ordinary rule-conformity is unchanged, but the verdict of $V_{\text{fmt}}^{(k)}$ flips. Hence clause (i) of theorem A.18 fails.

For (b), fix a displayed package $P = (T, B, R, S, I)$ with support basis $B = (b_1, \dots, b_n)$ and a nontrivial permutation π of $\{1, \dots, n\}$. Let V_{sup}^π return a positive verdict exactly when the underlying trace is rule-conformant and the displayed support basis occurs in the privileged order $(b_{\pi(1)}, \dots, b_{\pi(n)})$. Reordering the support basis while preserving its extensional members and the same displayed derivation target is a package-preserving reformulation at the level of constructional content, but the verdict of V_{sup}^π flips solely because support order has been privileged without re-admission. Hence either clause (i) or clause (ii) of theorem A.18 fails.

For (c), context-sensitive retargeting conventions fail clause (ii) of theorem A.18, because the same displayed package is treated as licensing a different target without explicit re-admission. For (d), support-basis substitutions not marked by re-admission likewise constitute support drift and so fail clause (ii). For (e), downstream-insensitive verdict mechanisms fail clause (iii) by theorem A.14, since positive and negative verdicts leave all downstream uptake possibilities unchanged. Therefore each listed family fails constructional governance realization. $\square$

Proposition A.21 (Decomposition of weaker uptake competitors by package-governance constraint). *Any weaker competitor to constructional governance must weaken at least one of exactly four package-governance constraints:*

- (i) verdict stability under package-preserving reformulation;
- (ii) anti-drift for target, support, scope, identity, and verdict;
- (iii) anti-repair for same-package reinterpretation and silent redescription; or
- (iv) support fixation together with downstream sensitivity of verdict differences.

No weaker competitor can depart from constructional governance without weakening at least one of these four constraints.

Proof. By theorem A.18, constructional governance for a displayed package is realized only if verdicts are stable under package-preserving reformulation, package drift is excluded absent re-admission or reset, same-package reinterpretative repair and silent redescription are excluded, and verdict differences govern at least one downstream uptake possibility for the same package. Any purported weaker competitor must therefore fail at least one of these requirements. If none were weakened, the competitor would simply satisfy the realization criterion and would not be weaker. So every weaker competitor weakens at least one of exactly these four package-governance constraints. $\square$

Theorem A.22 (Exhaustion of minimal weaker uptake competitors). *The minimal structurally distinct weaker competitors to constructional governance are exactly the following four classes: trace-only uptake, drift-tolerant uptake, revision-tolerant same-package uptake, and support-flexible uptake. No fifth minimal weaker competitor remains.*

Proof. By theorem A.21, any weaker competitor must weaken at least one of exactly four package-governance constraints. Minimality then yields four and only four minimal structurally distinct competitors.

If verdict stability is weakened while no further drift or reinterpetive latitude is added, the verdict remains at the level of bare trace-classification; this is trace-only uptake. If anti-drift is weakened, verdict use may permit target, support, scope, identity, or verdict drift without re-admission; this is drift-tolerant uptake. If anti-repair is weakened while the same package is still treated as in play, later reinterpetive repair or same-package revision is allowed; this is revision-tolerant same-package uptake. If support fixation together with downstream sensitivity is weakened, the support basis may be treated as flexibly substitutable or verdict differences may fail to govern any downstream uptake possibility; this is support-flexible uptake. Any purported fifth minimal weaker competitor would have to weaken one of the four package-governance constraints already listed and so would collapse into one of these four classes. Hence no fifth minimal weaker competitor remains. $\square$

Theorem A.23 (Exclusion of weaker uptake forms). *The kernel does not attach to every form of syntactic uptake. The weaker uptake competitors isolated in theorem A.22 all fail to realize constructional governance and therefore do not carry the kernel.*

Proof. Trace-only uptake fails by theorem A.11: it leaves the verdict at the level of bare trace-classification. Drift-tolerant uptake fails by theorem A.19, since target, support, scope, identity, or verdict drift is permitted without re-admission. Revision-tolerant same-package uptake fails because later reinterpetive repair of what is still treated as the same package is exactly the silent-redescription route barred by clause (iv) of theorem A.18. Support-flexible uptake fails because support substitutions not marked by re-admission constitute support drift. By theorem A.22, these are the minimal weaker competitors. Since each fails constructional governance, the realization of governance at Level 2 is reached by elimination rather than by a stipulative definition of uptake. $\square$

Theorem A.24 (Level 2/Level 3 separation). *Level 2 constructional governance and Level 3 closure-stable certification are distinct. Level 2 is realized when a verdict mechanism governs downstream uptake for a displayed package under theorem A.18. Level 3 is realized only when that governed package also satisfies the stronger retained-lineage and no-deferred-repair constraints needed for closure-stable certification. Thus Level 3 strengthens Level 2; it does not first create it.*

Proof. By theorem A.18, Level 2 is reached once downstream uptake is stably governed for a displayed package under package-preserving reformulation, anti-drift, trajectory sensitivity, and no silent redescription. But theorems A.29 and A.32 already show that provisional constructional governance need not settle closure-stable certification. Conversely, closure-stable certification presupposes Level 2 because retained-lineage hardening strengthens an already governed package rather than first creating governance out of bare traces. Therefore the levels are distinct and ordered. $\square$

Corollary A.25 (Not all syntactic gates qualify for constructional governance). *A syntactic derivation-check counts as governance-bearing only when its use as constructional governance satisfies*

the presentation-package stability conditions of theorem A.18. Otherwise it is a trace-classifier or drift-tolerant bookkeeping and not a generator of construction-governed acts.

Proof. By theorem A.12, there exist syntactic gates that fail the realization test. By theorem A.18, failure of that test prevents constructional governance. $\square$

Proposition A.26 (Characterization of constructional governance). *A verdict mechanism counts as governance-bearing for a displayed package if and only if:*

- (i) it stably partitions licit and unlicensed downstream uptake possibilities for that displayed package;*
- (ii) the partition is preserved under package-preserving reformulation;*
- (iii) target, support, scope, identity, and verdict drift, together with silent reinterpretation or same-package repair, are excluded absent explicit re-admission or reset; and*
- (iv) positive and negative verdicts alter at least one downstream uptake possibility for that package.*

Equivalently, constructional governance is the stable licit/unlicensed control of downstream use for displayed packages.

Proof. The forward direction is just the content of theorem A.18: package-preserving verdict stability gives (i) and (ii), anti-drift and anti-repair give (iii), and trajectory sensitivity gives (iv). Conversely, if (i)–(iv) hold, then the verdict mechanism does more than accompany a displayed package or a list of candidate uptake possibilities: it fixes which uptake profiles are licit for that very package, does so stably under package-preserving reformulation, forbids reinterpretive drift and repair, and makes positive/negative verdict differences matter downstream. That is precisely the role filled by constructional governance in Appendix A. $\square$

Remark A.27 (Why this is not stipulation). The preceding theorem should not be read as a verbal re-description of syntax. Its content is eliminative: bare trace-checkability, displayed-package identity, displayed uptake-space, drift-tolerant verdicts, same-package repair, and weaker uptake competitors have all been ruled out beforehand. The characterization theorem therefore names the strongest verdict-regime left standing after those exclusions, rather than stipulating a privileged vocabulary for proof practice.

This appendix is eliminative, not reductive. It does not claim a substrate more primitive than displayed-package use from which governance is recovered by reduction. It claims instead that, once weaker verdict-regimes are excluded, what remains is constructional governance in the exact sense characterized above.

Lemma A.28 (Syntactic licensing does not eliminate standing; under the criterion it instantiates it). *If, in a formal calculus, syntactic rule-conformity is what makes a step count as licit constructionally, and if that licensed/unlicensed use satisfies theorem A.18, then standing is not absent. Rather, standing is realized extensionally by that very licit-status condition. Therefore “standing is just rule-licensed stephood” is not a counterexample to standing but an identification of its local realization under admissibility-invariant use.*

Proof. By theorem A.10, syntax alone is insufficient for standing. But if a given calculus takes rule-conformity to be exactly the condition under which a line counts as constructionally licit,

and if that licit/unlicensed use passes the governance realization test of theorem A.18, then the calculus has not eliminated standing; it has first realized constructional governance by a verdict that stably governs downstream uptake for the same package, and standing is realized there as the licit/unlicensed role in that inferential architecture. What is realized is a role in use, not a second post-syntactic event. $\square$

Theorem A.29 (Provisional governance, social grouping, and closure-stable certification are distinct). *Provisional constructional governance, social or historical grouping of proof episodes, and closure-stable certification are pairwise non-identical. In particular:*

- (i) *a draft, seminar sketch, blackboard argument, or early posting may already be construction-governed—that is, critiqued, decomposed, or provisionally reused—without thereby being closure-stable or finally certified;*
- (ii) *a community may socially group multiple stages under one heading (“the proof of theorem T”) without thereby preserving governance-valid act identity across those stages; and*
- (iii) *closure-stable certification concerns whether the act, as performed at its own act-time and throughout retained-lineage continuation, was already licensed under lawful-redescription and no-silent-redescription constraints.*

Proof. Provisional constructional governance is stronger than bare trace-assessment because the item is already being used to support critique, decomposition, or defeasible downstream reasoning. But it is weaker than closure-stable certification, since later clarification, supplementation, or replacement may still be needed before the act’s validity is fixed without remainder. Social or historical grouping is coarser still: it tracks theorem-lineage, archival continuity, authorship, or communal reference under one heading. Closure-stable certification asks a narrower question: whether the act, as performed at its own act-time, was already licensed in the repair-sensitive sense relevant to the present paper. Therefore the three notions are distinct. $\square$

Theorem A.30 (Kernel attachment at Level 2 constructional governance). *The kernel $K = \{\text{Adm}, \text{Stand}, \text{Ref}, \text{Irr}\}$ attaches already at Level 2 constructional governance, but only after weaker uptake forms have been excluded. More precisely: once a verdict mechanism stably governs which downstream uptake profiles are licit for a displayed package and survives the exclusions of theorem A.23, determinate reference, standing, irreversibility, and the admissibility boundary governing their joint enforceability are already in play. Level 3 adds only the stronger closure-stable certification constraints needed for retained-lineage hardening; it does not introduce the kernel for the first time.*

Proof. By theorem A.23, weaker uptake forms do not realize constructional governance and therefore do not carry the kernel. What remains is the Level 2 case in which a verdict mechanism governs downstream uptake for a fixed displayed package and satisfies theorem A.18. At that point package drift and silent reinterpretation are excluded, so determinate reference is already at issue. Because the verdict now governs licit/unlicensed downstream uptake, standing is already at issue. Because reinterpretive same-package repair is excluded, irreversibility is already live. Admissibility is then the governance boundary under which those conditions are jointly enforceable. So the kernel attaches at Level 2 by elimination of weaker uptake forms, not by stipulation. By theorem A.24, Level 3 contributes stricter retained-lineage and closure-stability conditions, but not the first appearance of the kernel. $\square$

Remark A.31 (Seminar and blackboard practice). Seminar sketches, blackboard arguments, and exploratory proof attempts may fall inside the present target when they are construction-governed under admissibility-invariant use. What is excluded is not provisionality as such, but silent redescription, role-shift, and repair-by-reinterpretation.

Theorem A.32 (Provisional constructional governance does not entail closure-stable certification). *An act may be construction-governed at Level 2 of theorem A.6 without yet being closure-stable at Level 3. Provisional constructional governance is therefore stronger than bare trace-assessment but weaker than closure-stable certification.*

Proof. By definition, constructional governance already requires that the act be used to license assertion, rejection, critique, provisional reuse, or downstream extension under the realization criterion. So it exceeds bare trace-assessment and already carries the kernel roles at Level 2. But by theorem A.29, such provisional governance need not settle that the act was already closure-stable at act-time and throughout retained-lineage continuation. Therefore constructional governance and closure-stable certification are distinct levels. $\square$

Remark A.33 (Syntax is useful but not standing as such). The claim of this appendix is not that syntax is dispensable or always merely bookkeeping. It is that syntax is not standing *as such*. A syntactic verdict counts as governance-bearing only under the presentation-package stability constraints of theorem A.18; otherwise it remains a presentation-relative classifier of traces.

Lemma A.34 (Proof-system uptake trichotomy). *Let Σ be a formal calculus equipped with derivation-checking. Then exactly one of the following holds:*

- (i) Σ is studied only at the level of bare traces, in which case it is not yet construction-governed reasoning for the present target;
- (ii) Σ is construction-governed and its verdict mechanism survives the package-stability and weaker-uptake exclusion theorems, so that a verdict mechanism can function within constructional governance for displayed packages only after the package-stability and weaker-uptake exclusion tests have been passed; syntax alone never suffices; or
- (iii) Σ permits later reclassification, repair-by-reinterpretation, or silent redescription of constructional status, in which case irreversibility becomes a live issue rather than an eliminated one.

So no formal calculus supplies a counterexample merely by exhibiting derivation-checkability. The alleged fourth case in which “syntax just is the full governance role” without qualification is absorbed into case (ii) only when the verdict passes the presentation-package stability test of theorem A.18; when it does not, the case collapses into trace-classification or reinterpretive drift.

Proof. If Σ is not used to license assertion, rejection, critique, provisional reuse, or downstream extension, we are in case (i) by theorem A.11. If it is so used, theorem A.15 yields bridge-directed constructional governance. When the constructional use of its verdict satisfies theorem A.18 and survives theorem A.23, we are in case (ii): a verdict mechanism can function within constructional governance for displayed packages only after the presentation-package stability and weaker-uptake exclusion tests have been passed. Syntax alone is never sufficient by theorems A.10 and A.13. If instead constructional significance changes under later reinterpretation, role-shift, silent target/support drift, or verdict reclassification, then the criterion fails and irreversibility remains contested, giving case (iii). These cases are exhaustive. $\square$

Theorem A.35 (Inferential governance is not optional for proposition-bearing construction in the present sense). *Let Σ be a formal calculus. Suppose verdicts of Σ are used to license assertion, rejection, critique, provisional reuse, or downstream constructional extension. Then Σ is not functioning merely as a trace-classifier. It is being used as constructional governance at Level 2. If, moreover, that use satisfies theorem A.18, then constructional governance is realized there, and with it the standing/admissibility roles attach there as well, even when they are realized directly by the licensed syntactic verdict itself under the presentation-package stability test.*

Proof. If verdicts of Σ are used to license assertion, rejection, critique, provisional reuse, or downstream extension, then some derivation-traces are being treated as entitled for inferential deployment and others are not. That is already a licit/unlicensed distinction at the level of use, so Σ is not functioning merely as a trace-classifier. By theorem A.15, its use is bridge-directed. When that constructional use satisfies the invariance conditions of theorem A.18, the verdict counts as governance-bearing in use. By theorem A.30, the standing/admissibility roles attach there as well. So constructional governance is not optional decoration on top of reasoning; it is part of what makes the calculus function as proposition-bearing construction in the present sense. $\square$

Theorem A.36 (Trace-validity alone does not fix construction-governed act-status). *Derivation-checkability by itself does not determine construction-governed act-status. More precisely: from the fact that a finite sequence is rule-conformant in a formal calculus, it does not yet follow, without admissibility-invariant constructional use, (i) that the sequence functions as a claim-bearing act, (ii) that its conclusion is licensed for critique, provisional reuse, or downstream extension, or (iii) that later correction-sensitive validity questions are settled at act-time.*

Proof. A derivation-check returns only a presentation-relative verdict of conformity to a stated calculus. From that verdict alone one obtains, at most, that the sequence matches formation and rule-schema constraints internal to the presentation. Whether that verdict is being used to authorize a claim, bind a conclusion for later critique or reuse, or distinguish valid continuation from later rescue is not fixed by rule-conformity alone. By theorems A.10 and A.18, those stronger statuses arise only under admissibility-invariant constructional use. Therefore trace-validity alone does not determine construction-governed act-status. $\square$

Theorem A.37 (Act-time validity is stricter than theorem-lineage validity). *A theorem may remain historically supported through later correction, supplementation, or replacement while the original act that first purported to prove it remains invalid at its own act-time. Therefore theorem-lineage preservation does not entail governance-valid act preservation.*

Proof. Let a theorem-claim T be associated historically with an original proof-attempt π and a later corrected proof π' . It may be true that the mathematical community continues to group π and π' together under the heading “the proof of T ”. But this historical grouping tracks theorem-lineage, not whether π was already licensed at the time it was performed. If π' supplies material not available to π at act-time, then the validity of π' does not retroactively make π valid as the same act. Any contrary claim would require an untracked reinterpretation of what the original act already licensed, i.e. a silent redescription in the sense of theorem A.9. So theorem-lineage preservation is strictly weaker than governance-valid act preservation. $\square$

Theorem A.38 (Same-act repair and lineage-successor correction are disjoint). *Any later correction that changes the material required for validity either (i) leaves the original act invalid, or (ii) yields a*

governance-distinct successor act. No third possibility preserves both original invalidity and same-act validity.

Proof. Suppose a later correction changes the material required for validity. If the original act remains exactly the same governance-valid act, then its validity would have to have been fixed at act-time without the later material, contradicting the hypothesis that the later correction was needed. If one instead claims that the original act becomes valid by reinterpretation of what it always licensed, then one is invoking silent redescription of reference, role-assignment, or admissibility classification, contrary to theorem A.9. So either the original act remains invalid, or the corrected item is a distinct governance-valid successor. There is no third case in which an act both needed later correction and was already valid as that same act. $\square$

Remark A.39 (Historical grouping and act-time validity). Historical grouping of proof-attempts tracks theorem-lineage or community memory, not necessarily governance-valid act-status at act-time. Later correction may preserve theorem-lineage while failing to preserve same-act validity.

Lemma A.40 (Reference is necessary for determinate propositional output). *If a purported construction act lacks determinate reference, then it does not yield a determinate proposition of the regime.*

Proof. A proposition must be propositionally about something in a fixed way in order to bear governed constructional status. If the act's expressions refer equivocally or vacuously, then no single governed content is fixed: different readings, or none at all, remain equally compatible with the same mark sequence. What remains is a string, not a determinate proposition. $\square$

Lemma A.41 (Standing is necessary for derivational stephood). *If a purported step lacks standing, it does not function as a step in reasoning, but only as an unsupported inscription or transition.*

Proof. A derivational step is not merely something that occurs between two inscriptions; it purports to advance a claim under some licensed constructional status. Without standing there is no licit claim-act whose constructional role can be assessed. The item may still be written or computed, and may even be certified as rule-conforming inside a calculus, but by theorems A.11 and A.28 that is either bare trace classification or the local realization of standing rather than its elimination. So absent standing there is no reasoning step in the present sense. $\square$

Lemma A.42 (Irreversibility is necessary for stable derivational status). *If invalid steps are repairable post hoc without loss, then the distinction between valid derivation and retrospective rescue is not stably fixed at act-time.*

Proof. Suppose invalidity could later be removed while preserving the same act as fully valid. Then whether a purported step counted as licensed when performed would remain hostage to later repair. But a derivation whose status is not fixed at act-time has no stable difference between continuation and rescue. Its validity conditions therefore collapse into retrospective redescription. Ordinary mathematical compression of an original proof and a later corrigendum into “the same proof” does not refute this, because by theorems A.8 and A.29 historical identity is coarser than governance-valid act identity. $\square$

Lemma A.43 (Admissibility is the governance boundary of joint enforceability). *Whenever determinate reference, standing, and irreversibility are jointly enforceable on the same construction acts, admissibility is already in force there as the governance boundary of that joint enforceability.*

Proof. If determinate reference, standing, and irreversibility are jointly enforceable on the same acts, then the regime already distinguishes licensed from non-licensed claim-acts and distinguishes valid continuation from retrospective rescue. That governance boundary is exactly what admissibility names. This need not be an extra mechanism layered over syntax: in a formal calculus with constructional governance under admissibility-invariant use it may be realized by the licit/unlicensed role played by rule-governed derivability, but only when that use satisfies the realization criterion of theorem A.18. Conversely, a regime lacking that boundary cannot jointly enforce the three conditions on the same acts. $\square$

Theorem A.44 (Assessability of bare derivation objects). *Bare derivation objects are assessable under weaker conditions than construction-governed reasoning acts under admissibility-invariant use. In particular, presentation-relative rule-conformity of finite traces may be checked without thereby fixing claim-status, inferential entitlement, or repair-sensitive act-validity.*

Proof. This is exactly the distinction established above. A derivation-check settles only whether a trace conforms to the presentation-level rules of a calculus. By theorem A.36, that does not yet settle whether the checked item functions as a claim-bearing construction act, whether its conclusion is licensed for critique, provisional reuse, or downstream extension, or whether its validity is fixed at act-time in the repair-sensitive sense relevant to governed derivation. Therefore bare trace-assessability is strictly weaker than construction-act assessability. $\square$

Theorem A.45 (Minimal assessability boundary for construction-governed acts under admissibility-invariant use). *Let R be any system whose outputs are treated not merely as checked traces but as construction-governed proposition-bearing construction acts under admissibility-invariant use, including assertion, rejection, critique, provisional deployment, provisional reuse, or downstream constructional extension. Then determinate reference, claim-bearing standing, and non-repairable invalidity are jointly necessary and minimal for that evaluability. More exactly:*

- (i) *if determinate reference is removed, proposition identity is lost and the outputs are no longer evaluable as determinate proposition-bearing content;*
- (ii) *if standing is removed, no transition qualifies as a licit claim-bearing step and the outputs are no longer evaluable as construction-governed acts under admissibility-invariant use;*
- (iii) *if irreversibility is removed, there is no stable distinction between valid continuation and retrospective rescue at act-time; and*
- (iv) *if an additional condition is imposed as an independent governance requirement beyond those three, then the regime is governed more tightly than construction-governed evaluability itself requires.*

Accordingly, determinate reference, standing, and irreversibility mark the minimal assessability boundary for construction-governed acts under presentation-stable admissibility-invariant use. The kernel therefore attaches already at Level 2 constructional governance; closure-stable certification adds stronger retained-lineage constraints without moving the attachment point.

Proof. Item (i) is theorem A.40. Item (ii) is theorem A.41, now read in light of theorems A.10, A.18, A.35 and A.36: trace-checkability is insufficient, and constructional governance requires a licit/unlicensed distinction that survives admissibility-invariant use. Item (iii) follows from

theorems A.37, A.38 and A.42: theorem-lineage correction does not settle governance-valid act-status retroactively, and same-act repair is blocked once silent redescription is excluded. For minimality, suppose an additional condition Q is imposed as an independent governance requirement on every construction-governed construction act under admissibility-invariant use. If Q is already enforced by the joint assessability work of reference, standing, and irreversibility, then it is not additional. If it excludes some act-sequences that already satisfy those three conditions, then it contributes further governance beyond what construction-governed evaluability itself requires. So the three conditions are jointly necessary and minimal. $\square$

Remark A.46. Theorems A.44 and A.45 block the formalist equivocation between two different assessment targets: the checkability of derivation-objects and the evaluability of construction-governed acts under admissibility-invariant use. The present paper concerns the second, because closure, critique, reuse, extension, and same-domain threat are questions about constructional behavior rather than about trace classification alone.

Corollary A.47 (Non-arbitrariness of the target class). *The class of meaningful regimes is not defined by pre-selecting the kernel. It is exactly the class of systems in which construction-governed determinate construction under presentation-stable admissibility-invariant use is assessable as reasoning for the present target. Restriction to that class is therefore not protective but forced by the target of inquiry.*

Proof. By theorem A.45, determinate reference, standing, and irreversibility are not optional entry criteria appended in advance of the argument. They are the minimal conditions under which construction-governed proposition-bearing outputs and transitions are evaluable as construction acts under admissibility-invariant use at all, whereas bare trace-assessment remains a weaker phenomenon by theorem A.44. So to restrict attention to regimes in which those conditions are assessable is simply to restrict attention to the subject matter of the paper rather than to protect the thesis from counterexample by fiat. $\square$

Theorem A.48 (Non-degenerate construction forces the kernel). *If a meaningful regime admits construction-governed non-degenerate construction under admissibility-invariant use, then admissibility, standing, reference, and irreversibility are already in force in that regime.*

Proof. Let R be a meaningful regime admitting construction-governed non-degenerate construction under admissibility-invariant use. By theorem A.45, the target here is not bare trace-checkability but construction-governed construction-act assessability under admissibility-invariant use. So any regime in which outputs are assessable as construction-governed determinate construction acts already requires determinate reference, claim-bearing standing, and irreversibility as the minimal assessability boundary of that evaluability. By theorem A.43, the joint enforceability of those conditions already places admissibility in force as the governance boundary constituted by their coherence on the same acts. Hence R already enforces the full kernel K . $\square$

Corollary A.49 (No non-degenerate regime below the kernel). *No meaningful regime can both support construction-governed non-degenerate construction under admissibility-invariant use and generate any member of K from a strictly lower governance-free basis within the same meaningful regime.*

Proof. Assume some meaningful regime supports construction-governed non-degenerate construction under admissibility-invariant use and nevertheless generates some $P \in K$ from a strictly lower

governance-free basis. By theorem A.48, the full kernel K is already in force in that regime before any such generation begins. So the alleged generator does not produce P from below; it operates only within a regime whose meaningfulness already depends on P . $\square$

Lemma A.50 (Derivation presupposes admissibility). *If a governance-valid derivation of any proposition P exists, then admissibility is already in force.*

Proof. By definition, a governance-valid derivation is a licit construction act-sequence rather than a bare formal trace. Its steps therefore count as steps only under an admissibility regime. $\square$

Lemma A.51 (Derivation presupposes standing). *If a governance-valid derivation of any proposition P exists, then standing is already in force.*

Proof. A derivation with no standing would be a sequence of acts with no licit claim-bearing status. Such a sequence might be written down, but it would not count as a derivation of anything. $\square$

Lemma A.52 (Derivation presupposes reference). *If a governance-valid derivation of any proposition P exists, then determinate reference is already in force.*

Proof. A derivation whose expressions fail to refer determinately does not derive a determinate conclusion. It supplies marks, not governed content. Since governance-valid derivation names a derivation of a proposition rather than a mere syntactic trail, reference is presupposed. $\square$

Lemma A.53 (Derivation presupposes irreversibility). *If a governance-valid derivation of any proposition P exists, then irreversibility is already in force.*

Proof. Without irreversibility, invalid steps could be repaired retroactively into acceptable ones, and the distinction between valid continuation and post hoc rescue would collapse. In that case the derivation would not possess a stable governance status at all. $\square$

Corollary A.54 (Global presupposition lemma). *For every proposition P , if there is a governance-valid derivation of P , then the full kernel K is already in force.*

Proof. Combine theorems A.50 to A.53. $\square$

Theorem A.55 (Kernel non-derivability). *No member of K is derivable from below.*

Proof. Let $P \in K$. Suppose, toward contradiction, that P is derivable from below. Then there exists a governance-valid derivation of P whose validity does not already presuppose P , any equivalent under identity-preserving transformation condition, or any condition whose own governance-validity already requires P . But by theorem A.54, every governance-valid derivation already presupposes the full kernel K , hence P itself. This is exactly the forbidden circularity in derivability from below. $\square$

Theorem A.56 (No lower generator / structural non-derivability). *Let $P \in K$. There exists no system or meaningful regime supporting construction-governed non-degenerate construction under admissibility-invariant use in which P is generated from any purportedly lower governance condition or package within the same meaningful regime. Equivalently, the non-derivability of P is structural, not merely a defect of one stipulated proof format.*

Proof. Assume, for contradiction, that some meaningful regime supports construction-governed non-degenerate construction under admissibility-invariant use and nevertheless generates some $P \in K$ from a purportedly lower governance package within the same meaningful regime. By theorem A.48, the full kernel K is already in force in that regime, and hence P is already in force there. So the alleged lower package does not generate P from below; it operates only inside a regime whose meaningfulness already depends on P . $\square$

Theorem A.57 (Proof-system uptake trichotomy). *Let Σ be any construction system, meta-system, or external presentation proposed as a candidate lower generator for some member of the kernel K . Then exactly one of the following holds:*

- (i) Σ functions only as a trace-classifier for derivation-objects;
- (ii) Σ is used with constructional governance, and its verdict mechanism survives the package-stability and weaker-uptake exclusion tests, so that it can function as constructional governance for displayed packages;
- (iii) Σ permits later reclassification, repair-by-reinterpretation, or silent redescription of act-status, in which case irreversibility remains live rather than eliminated.

No fourth candidate lower-generator type remains. In particular, the alleged case in which “syntax just is the full governance role” without qualification is absorbed into case (ii) when the realization criterion is satisfied and collapses into case (i) or case (iii) when it is not.

Proof. If Σ is not used to license assertion, rejection, critique, provisional reuse, or downstream extension, then it is only a trace-classifier by theorem A.36. If it is so used, then by theorem A.35 the system is being used as constructional governance rather than mere trace-classification. When that constructional use satisfies theorem A.18 and survives theorem A.23, the verdict mechanism can function as constructional governance for displayed packages, giving case (ii). If instead constructional significance may later be altered by reinterpretation, role-shift, or silent redescription while purporting to preserve the same governance-valid act, then by theorems A.37 and A.38 irreversibility has not been removed but made contested, giving case (iii). These cases are exhaustive. $\square$

Theorem A.58 (Regime-invariant non-derivability). *Let $P \in K$. There exists no derivation of P from any construction system, meta-system, or external presentation Σ that both*

- (i) *fails to have derivation-validity conditions already enforcing P , a equivalent under identity-preserving transformation condition, or a condition whose own derivation-validity already requires P , and*
- (ii) *nevertheless purports to support non-degenerate reasoning in the same meaningful regime.*

Changing formal dress, moving to a meta-theory, or recoding the language therefore does not remove the need for the governance work named by P .

Proof. Let $P \in K$, and suppose some system Σ is offered as a derivation system for P . By theorem A.57, every candidate Σ falls into one of three classes.

In class (i), Σ functions only as a trace-classifier. Then it is not yet a generator of construction-governed determinate construction acts under admissibility-invariant use, and so it cannot derive P from below within the same meaningful regime.

In class (ii), Σ is used with constructional governance and, by theorem A.18, already realizes the standing/admissibility roles extensionally in its licensed/unlicensed use. If the validity conditions under which a Σ -derivation counts as a derivation already enforce P , a equivalent under identity-preserving transformation condition, or a condition whose own validity already requires P , then Σ does not generate P from below. It begins under the very governance it purports to derive.

In class (iii), Σ permits later reclassification or repair-sensitive revision of act-status. Then by theorems A.37 and A.38 irreversibility has not been removed but remains live rather than eliminated.

So no candidate lower generator removes the need for the governance work named by P . A non-kernel-respecting Σ is, at best, an external coding, a meta-level bookkeeping device, or a purely formal trace; it is not a derivation of P from below within the same meaningful regime. $\square$

Theorem A.59 (Calculus plurality does not undercut role necessity). *Differences among Hilbert, natural-deduction, sequent, type-theoretic, or comparable formal presentations do not by themselves remove the need for claim-status, constructional governance, and repair-sensitive act-validity whenever those systems are used as construction-governed reasoning under admissibility-invariant use. Accordingly, presentation plurality does not supply a lower generator of the kernel.*

Proof. The relevant distinction is not which proof calculus is chosen, but whether the chosen calculus is being treated merely as a trace-classifier or as licensing construction-governed acts under admissibility-invariant use. By theorem A.57, every such calculus falls into trace-only use, extensional realization of the standing/admissibility roles only under the realization criterion, or a regime in which irreversibility remains live. Changing presentation therefore does not by itself remove the roles isolated in the kernel. $\square$

Proposition A.60 (Admissibility requires standing). *Any regime lacking standing is not an admissibility regime.*

Proof. A regime without standing does not distinguish licit claim-making from unsupported utterance. It therefore lacks the governance force required to count as admissibility. At best it describes transitions; it does not govern construction. $\square$

Proposition A.61 (Standing requires reference). *Standing without determinate reference is incoherent.*

Proof. Standing is the licitness of a claim. But where reference is indeterminate, there is no determinate claim whose licitness could be assessed. A purported standing relation with no reference relation underneath it licenses emptiness or equivocation, not governed construction. $\square$

Proposition A.62 (Governed reference requires irreversibility). *The reference relevant to non-degenerate reasoning is not bare denotation but reference fixed for governed constructional use at act-time. That form of reference cannot be maintained if invalid claim-acts are repairable post hoc without loss.*

Proof. A symbol may still bear a denotational relation under later reinterpretation or repair, but that is not yet the kernel notion of reference. If later repair can retroactively convert the same invalid act into a fully valid one without loss, then the constructional status under which the act's target is fixed is not settled at act-time. What remains may be denotation, but not governed reference in the sense required for non-degenerate construction. $\square$

Proposition A.63 (Governance irreversibility requires admissibility). *The irreversibility relevant here is not mere temporal one-wayness or procedural non-rewindability. It is the governance distinction between licensed continuation and prohibited repair, and that distinction is available only under an admissibility boundary.*

Proof. Without admissibility there is no principled governance boundary by which a continuation counts as licensed and a modification counts as illicit repair. One can still speak of sequence or alteration, but not of governance-irreversible invalidity. $\square$

Theorem A.64 (Mutual closure of the kernel). *The members of K form a mutually closed necessity set: none can be coherently asserted while denying the governance role of the others.*

Proof. Theorems A.60 to A.63 exhibit a governance-specific necessity loop: admissibility requires standing, standing requires reference, governed reference requires irreversibility, and governance irreversibility requires admissibility. The loop is not merely terminological. By theorem A.45, each member is required for construction-governed construction-act assessability under admissibility-invariant use, and by theorem A.57 apparent eliminations of one member either collapse to trace-only classification, realize the role only under the realization criterion, or leave the disputed condition still live. Hence no member can be detached from the rest while retaining its intended governance content. $\square$

Theorem A.65 (Minimality of the kernel). *The kernel K is minimal for non-degenerate reasoning: no proper subset of K suffices for non-degenerate reasoning.*

Proof. Assume a proper subset $K_0 \subsetneq K$ suffices for non-degenerate reasoning. A purported proper subset may still suffice for bare trace-checkability by theorem A.44, but that is not yet the target phenomenon. Let $P \in K \setminus K_0$ be an omitted member. Since construction-governed non-degenerate construction under admissibility-invariant use allegedly remains possible, either P is unnecessary or P is generated from the remaining members. The first option contradicts theorem A.64, since the governance role of each member depends on the rest. The second contradicts theorem A.55, since P would then be derivable from below from a lower basis. So for construction-governed acts, omission of a kernel member either destroys the target class or reintroduces the omitted role only under the realization criterion. Both possibilities fail. $\square$

Corollary A.66 (No faithful lower generator in the same meaningful regime). *There exists no faithful lower generator for admissibility, standing, reference, or irreversibility within the same meaningful regime; and no faithful same-domain extension reopens that possibility.*

Proof. The internal claim is theorem A.56. The faithful-extension-stable claim is theorem A.58. Together with theorem A.65, they show that primitive status here means structural non-derivability within the same fixed-domain meaningful regime rather than merely a presentational decision to stop derivation. $\square$

Remark A.67 (Role of Appendix A). Appendix A no longer functions as background staging. It proves the necessity floor for the target class itself. In particular it does not concede a counterexample from Hilbert-style, natural-deduction, type-theoretic, or comparable calculi merely because those calculi admit mechanical proof-checking: the appendix distinguishes bare trace classification from constructional governance and closure-stable certification, shows that constructional governance realizes the standing/admissibility roles only under the presentation-package stability governance

realization theorem, separates historical proof identity from governance-valid act identity, and proves that presentation plurality does not supply a lower generator of the kernel. Every later appendix and every theorem in the main body works downstream of the in-paper facts that construction-governed determinate construction under presentation-stable admissibility-invariant use has a minimal assessability boundary, that non-degenerate construction forces the full kernel, and that no faithful same-regime or faithful same-domain extension derives that kernel from below.

B Interface forcing: AMetric boundary, bivalence, standing emergence, and conservation

This appendix derives the interface consequences for the original governing use. Kernel necessity is the fixed starting point. The equality-only AMetric results retain their stated input, while complete original-answer semantics supplies the downstream conservation, quotient, and interior results. Binary qualification is distinguished from the full content needed for its original uses.

B.1 The full symmetry-based AMetric boundary package

Definition B.1 (Pre-admissible proposal space). A *pre-admissible proposal space* is a nonempty set P considered prior to admissibility fixation. At this layer no names, coordinates, indices, representatives, or privileged charts carry authority. Equality is the only authoritative structure on P .

Definition B.2 (Nontrivial proposal regime). The proposal regime is *nontrivial* if $|P| \geq 2$. When $|P| = 1$, all impossibility claims below become vacuous.

Theorem B.3 (Label-neutrality forces full relabeling symmetry). *For any pre-admissible proposal regime satisfying the preceding definitions, every bijection $\pi : P \rightarrow P$ preserves all admissibility-relevant structure available on the pre-admissible side. Hence the admissibility-relevant symmetry group of the pre-boundary presentation is the full symmetric group $\text{Sym}(P)$.*

Proof. By definition, equality is the only authoritative pre-admissible structure. Every bijection preserves and reflects equality, hence is an automorphism of the full pre-boundary authoritative structure. No proper subgroup can be singled out without adding further admissibility-relevant authority in names, coordinates, representatives, charts, or selectors, exactly what the definition forbids. $\square$

Definition B.4 (Relabeling symmetry). By theorem B.3, the relevant symmetry group on the pre-admissible side is the full symmetric group $\text{Sym}(P)$ of bijections $\pi : P \rightarrow P$. Each $\pi \in \text{Sym}(P)$ acts componentwise on tuples by

$$\pi(x_1, \dots, x_n) := (\pi(x_1), \dots, \pi(x_n)).$$

Definition B.5 (Pre-admissibly meaningful relation). For $n \geq 1$, an n -ary relation $R \subseteq P^n$ is *pre-admissibly meaningful* if it is invariant under relabeling: for every $\pi \in \text{Sym}(P)$ and every $\bar{x} \in P^n$,

$$\bar{x} \in R \iff \pi(\bar{x}) \in R.$$

Likewise, a map $f : P \rightarrow A$ is meaningful when $f \circ \pi = f$ for all $\pi \in \text{Sym}(P)$, and a selector $c : \mathcal{P}(P) \setminus \{\emptyset\} \rightarrow P$ is meaningful when it is equivariant:

$$c(\pi(A)) = \pi(c(A)) \quad \text{for all } \pi \in \text{Sym}(P), A \neq \emptyset.$$

Proposition B.6 (Invariance is the minimal criterion of pre-boundary authority). *On a proposal side satisfying the preceding definitions, any admissibility-relevant relation, grading, or selection rule must be preserved under relabelings that alter only labels and not proposal identity as proposals. Accordingly, symmetry-invariance is not an added semantic filter but the minimal formal expression of pre-boundary admissibility relevance.*

Proof. Suppose an admissibility-relevant rule were not preserved under such a relabeling. Then two presentations differing only by labels would receive different pre-boundary authority. That would make the labels, coordinates, or representatives themselves admissibility-relevant, contrary to the definition of the proposal space. So invariance is required before any stronger structural claim is made. $\square$

Definition B.7 (Equality pattern). For a tuple $\bar{x} = (x_1, \dots, x_n) \in P^n$, the *equality pattern* of $\bar{x}$ is the partition of $\{1, \dots, n\}$ whose blocks are the index sets of equal coordinates. Equivalently, $\bar{x}$ and $\bar{y}$ have the same equality pattern iff

$$x_i = x_j \iff y_i = y_j \quad \text{for all } i, j \in \{1, \dots, n\}.$$

Lemma B.8 (Transitivity and two-transitivity of $\text{Sym}(P)$). *Assume the proposal regime is nontrivial. Then:*

- (i) *for any $x, y \in P$ there exists $\pi \in \text{Sym}(P)$ with $\pi(x) = y$;*
- (ii) *for any distinct $x, y, u, v \in P$ with $x \neq y$ and $u \neq v$, there exists $\pi \in \text{Sym}(P)$ with $\pi(x) = u$ and $\pi(y) = v$.*

Proof. Part (i) is immediate: any bijection sending x to y and acting arbitrarily on the rest of P lies in $\text{Sym}(P)$. For part (ii), define a bijection on the two-point subset $\{x, y\}$ by $x \mapsto u$ and $y \mapsto v$; because the points are distinct, the partial map extends to a bijection of all of P . $\square$

Lemma B.9 (Orbit classification by equality pattern). *Fix $n \geq 1$. Two tuples $\bar{x}, \bar{y} \in P^n$ lie in the same $\text{Sym}(P)$ -orbit if and only if they have the same equality pattern.*

Proof. If $\bar{y} = \pi(\bar{x})$ for some $\pi \in \text{Sym}(P)$, then injectivity of π preserves and reflects equality, so the equality patterns coincide. Conversely, assume the equality patterns agree. List the distinct values appearing among the coordinates of $\bar{x}$ as $a_1, \dots, a_k$. Because the equality patterns coincide, the coordinates of $\bar{y}$ also use exactly k distinct values, say $b_1, \dots, b_k$, in matching positions. The partial map $a_r \mapsto b_r$ is therefore well-defined and injective, and it extends to a bijection $\pi : P \rightarrow P$ with $\pi(\bar{x}) = \bar{y}$. $\square$

Theorem B.10 (Equality-pattern completeness). *Every pre-admissibly meaningful relation $R \subseteq P^n$ is completely determined by equality patterns. Equivalently, R is a union of $\text{Sym}(P)$ -orbits, and those orbits are exactly the equality-pattern classes from theorem B.9.*

Proof. Let $\bar{x}, \bar{y} \in P^n$ have the same equality pattern. By theorem B.9, there is $\pi \in \text{Sym}(P)$ with $\pi(\bar{x}) = \bar{y}$. If R is meaningful, then

$$\bar{x} \in R \iff \pi(\bar{x}) \in R \iff \bar{y} \in R.$$

So membership depends only on the equality pattern. The converse is immediate because relabelings preserve equality patterns. $\square$

Corollary B.11 (Unary collapse). *Every meaningful unary predicate $U \subseteq P$ is trivial: either $U = \emptyset$ or $U = P$. Equivalently, every meaningful map $f : P \rightarrow A$ is constant.*

Proof. For $n = 1$ there is only one equality pattern. So theorem B.10 forces every meaningful unary predicate to be constant on all of P . The map claim follows because the fibers of a meaningful map are meaningful unary predicates. $\square$

Proposition B.12 (No nonconstant meaningful grading). *Let A be any set with $|A| \geq 2$. Every meaningful map $f : P \rightarrow A$ is constant. In particular, there is no admissibility-relevant score, index, rank, or label on individual proposals.*

Proof. This is the map form of theorem B.11. For any $x, y \in P$, theorem B.8(i) yields a relabeling π with $\pi(x) = y$. Meaningfulness gives $f(y) = f(\pi(x)) = f(x)$. $\square$

Proposition B.13 (No nontrivial meaningful strict order). *Let $R \subseteq P^2$ be a meaningful relation that is asymmetric. Then R contains no ordered pair of distinct elements. In particular, there is no meaningful strict total order, strict weak order, or other admissibility-relevant asymmetric ranking on P .*

Proof. Suppose, toward contradiction, that xRy for some distinct $x, y \in P$. Let π be the transposition swapping x and y . Because R is meaningful and $\pi(x, y) = (y, x)$,

$$xRy \iff yRx,$$

contradicting asymmetry. $\square$

Proposition B.14 (No equivariant selector). *There is no meaningful selector*

$$c : \mathcal{P}(P) \setminus \{\emptyset\} \rightarrow P$$

with $c(A) \in A$ for every nonempty A .

Proof. Assume such a selector exists. Choose distinct $x, y \in P$ and set $A = \{x, y\}$. Let π be the transposition swapping x and y . Then $\pi(A) = A$, so equivariance gives

$$c(A) = c(\pi(A)) = \pi(c(A)).$$

But $c(A)$ is either x or y , and the transposition sends each of those to the other. So $c(A) \neq \pi(c(A))$, contradiction. $\square$

Theorem B.15 (Metric collapse). *Let $d : P \times P \rightarrow \mathbb{R}_{\geq 0}$ be a meaningful metric. Then there exists a constant $\lambda > 0$ such that*

$$d(x, x) = 0 \quad \text{and} \quad d(x, y) = \lambda \text{ for all } x \neq y.$$

So every meaningful metric is the discrete equality metric up to overall scale.

Proof. Because d is a metric, $d(x, x) = 0$ for every x . Now take any distinct pairs (x, y) and (u, v) . By theorem B.8(ii), there is $\pi \in \text{Sym}(P)$ with $\pi(x) = u$ and $\pi(y) = v$. Meaningfulness gives

$$d(u, v) = d(\pi(x), \pi(y)) = d(x, y).$$

Hence every distance between distinct points has the same value, say λ . Positivity of the metric gives $\lambda > 0$. $\square$

Theorem B.16 (AMetric boundary). *Assume a nontrivial pre-admissible proposal regime. Then every pre-admissibly meaningful relation on P^n is determined solely by equality patterns. Consequently:*

- (A1) *there is no nonconstant meaningful grading or indexing of individual proposals;*
- (A2) *there is no meaningful asymmetric ranking of distinct proposals;*
- (A3) *there is no meaningful selector choosing a canonical representative from a nontrivial set of proposals; and*
- (A4) *every meaningful metric collapses to the discrete equality metric and therefore contributes no graded admissibility content.*

In this precise sense the pre-admissible proposal regime is AMetric and non-parameterized.

Proof. The equality-pattern statement is theorem B.10. Items (A1)–(A4) are exactly theorems B.12 to B.15. □

Corollary B.17 (No admissibility-relevant parameterization). *There is no admissibility-relevant continuous or discrete parameter on the pre-admissible side that distinguishes multiple positive ways of crossing the boundary. Any purported such parameter either collapses to bookkeeping, reduces to trivial equality information, or imports post-fixation structure.*

Proof. A continuous or discrete parameter on individual proposals would be a meaningful map or grading and is ruled out by theorem B.12. A parameter based on distance or approach direction is ruled out by theorem B.15. A parameter based on selecting one candidate from many is ruled out by theorem B.14. □

B.2 No deeper pre-boundary invariant and failure-class exhaustion

Definition B.18 (Purported deeper invariant). *A purported deeper invariant is a family of pre-admissibly meaningful relations on P claimed to determine or constrain admissibility more fundamentally than the boundary classification itself. The claim of “deeper” means that the invariant is supposed to be operative before admissibility is fixed and yet strong enough to decide, or partly decide, which proposals qualify.*

Lemma B.19 (Constraint requires pre-boundary availability). *If a purported deeper invariant really constrains boundary classification, then it must be available and meaningful on the pre-admissible side. A purely post-fixation invariant cannot ground admissibility without circularity.*

Proof. If an invariant is available only after admissibility is already fixed, then using it to determine admissibility presupposes the very status under determination. So any genuinely constraining deeper invariant must already be available pre-admissibly. □

Lemma B.20 (Invariant data induce invariant unary criteria). *Let Q be any collection of pre-admissibly meaningful relations on P . Any criterion $U(x)$ for single proposals that is defined solely from Q , without adding non-invariant constants, labels, or privileged representatives, is itself a meaningful unary predicate on P .*

Proof. The defining rule for U uses only equality, logical operations, quantification over P , and membership in the relations from Q . Because each relation in Q is meaningful, its truth values are preserved under relabeling, and equality is preserved as well. So for every $\pi \in \text{Sym}(P)$ one has $U(x) \Leftrightarrow U(\pi(x))$. $\square$

Theorem B.21 (No deeper pre-boundary invariant with independent constraining force). *Let Q be a purported deeper invariant. Then exactly one of the following holds:*

- (D1) *Q relies on non-invariant labels, coordinates, selectors, or post-fixation structure, hence is an illicit import;*
- (D2) *every unary admissibility criterion induced from Q is trivial, so Q has no independent constraining force on individual proposals; or*
- (D3) *once admissibility is fixed, the only nontrivial constraining role played by Q is extensionally identical to the boundary classification already being determined.*

There is no fourth option.

Proof. By theorem B.19, any genuinely deeper invariant must be available pre-admissibly. If it is not meaningful under relabeling symmetry, then by theorem B.6 it assigns admissibility-relevant authority to label changes alone, so it uses extra structure and we are in case (D1). So assume Q is meaningful. Any way of turning Q into a criterion for whether an individual proposal should pass the boundary produces, by theorem B.20, a meaningful unary predicate on P . By theorem B.11, every such unary predicate is trivial. That yields (D2) unless extra structure is imported. The only remaining possibility is post-fixation extensional coincidence with the boundary classification itself, which is (D3). $\square$

Definition B.22 (Failure classes). Relative to a fixed domain D , the possible structural failures are:

- F1. **Silent redescription:** eligibility changes under identity-preserving transformation without an explicit reset of target or criterion.
- F2. **Deferred validation:** an act enters licensed composition before its eligibility is fixed.
- F3. **Post-selection repair:** an ineligible evaluated act is later treated as eligible without fresh evaluation.
- F4. **Scope transport:** structure from a distinct domain or envelope is used to legalize an act on D .
- F5. **Parameter injection:** extra metric, probabilistic, scalar, or indexing structure is introduced to decide eligibility.

Lemma B.23 (Locus reduction for admissibility evasion). *Any attempted violation of fixed-domain licensed construction must alter at least one of the following:*

- (i) *redescription stability of admissibility-relevant verdicts;*
- (ii) *the temporal order between evaluation and entry into licensed composition;*
- (iii) *the identity-binding of a previously evaluated act under same-scope continuation;*
- (iv) *the scope or envelope of authority under which the verdict is claimed to hold; or*

- (v) the pre-boundary decision structure by introducing metric, ranking, selector, probabilistic, or comparable parameterization.

These loci correspond respectively to F1–F5 of theorem B.22.

Proof. Take any attempted evasion of a prior admissibility gate on a fixed domain. If identity-preserving transformation alone can change the verdict, locus (i) is altered. If entry into licensed composition occurs before evaluation is fixed, locus (ii) is altered. If a previously evaluated act is later treated as eligible as the same act without fresh evaluation, locus (iii) is altered. If authority is borrowed from a different domain or evaluative envelope, locus (iv) is altered. If none of (i)–(iv) is altered yet eligibility is still decided in a nontrivial way before fixation, then the only remaining place for the evasion to enter is added pre-boundary decision structure, namely locus (v). These five loci are exactly F1–F5. $\square$

Theorem B.24 (Failure-class exhaustion). *Every violation of reference-stable, irreversibility-respecting licensed construction instantiates at least one failure class in theorem B.22.*

Proof. By theorem B.23, any attempted violation must alter at least one of the five loci listed there. Those loci are exactly the failure classes F1–F5 from theorem B.22. So every violation instantiates at least one such class. $\square$

Theorem B.25 (Collapse under each failure class). *Each class in theorem B.22 destroys at least one of the following:*

- (a) stable reference under identity-preserving transformation;
- (b) well-defined licensed composition; or
- (c) irreversible commitment.

Proof. F1. Silent redescription directly violates stable reference under identity-preserving transformation.

F2. Deferred validation invokes licensing before the entry condition for licensing has been fixed. So composition is not well-defined as licensed composition.

F3. Post-selection repair either treats the same evaluated act as newly eligible, in which case earlier rejection is not binding and irreversibility is undermined, or it silently changes identity, which violates stable reference.

F4. Scope transport mixes envelopes with different admissibility conditions. That destroys either reference stability or the well-definedness of licensed composition on the target scope.

F5. Parameter injection contradicts theorem B.16 by importing pre-boundary ranking or metric structure, so the gate is no longer well-defined on the neutral proposal regime. $\square$

B.3 Gate necessity, same-side split collapse, and standing emergence

Definition B.26 (Admissibility predicate on a fixed domain). Fix a domain D with evaluated fragment $X_D^\bullet$. An admissibility predicate is a map

$$\text{Adm}_D : X_D^\bullet \rightarrow \{0, 1\}$$

read as admitted/not-admitted on the fixed scope.

Theorem B.27 (Necessity of a prior admissibility gate). *Under the kernel floor of Appendix A together with the pre-boundary symmetry package above, every non-degenerate same-domain construction regime induces an admissibility gate that is checked before licensed same-domain composition.*

Proof. Assume no such prior predicate exists. Then an act must enter licensed composition without evaluation, which is F2; or it must be legalized later by repair, which is F3; or it must be authorized by imported structure from elsewhere, which is F4; or it must be ranked or thresholded by injected parameters, which is F5. By theorem B.25, every route destroys one of the base goods the theorem assumes. Therefore a prior eligibility predicate is forced. $\square$

Definition B.28 (Status interface). A *status interface* on D is a pair $(\text{Status}_D, \text{Good}_D)$ where $\text{Status}_D : X_D^\bullet \rightarrow V_D$ assigns a status value to each evaluated act in its domain, and $\text{Good}_D \subseteq V_D$ designates those values treated as eligible for commitment, reuse, or standing-bearing continuation on D . Write $\text{Bad}_D := V_D \setminus \text{Good}_D$. The interface is *admissibility-relevant* when the statuses it distinguishes affect present eligibility or later standing-bearing continuation on the fixed domain.

Definition B.29 (Same-side split). A *same-side split* for $(\text{Status}_D, \text{Good}_D)$ is a pair of distinct values $u \neq v \in V_D$ that lie on the same side of the interface: either both $u, v \in \text{Good}_D$ or both $u, v \in \text{Bad}_D$.

Definition B.30 (Same-scope promotion). A *same-scope promotion* occurs when an act that has already received a status in Bad_D on the fixed domain D is later treated as eligible on that same domain without an explicit reset or scope change.

Lemma B.31 (No same-scope repair or promotion). *Under the kernel floor and same-domain scope discipline, no same-scope promotion is possible.*

Proof. Assume a same-scope promotion occurs. Then an act first treated as ineligible on D is later treated as eligible on the same fixed domain without explicit reset. There are only three possibilities. (i) Reinterpretation: the later stage treats the act as having meant something different from what it meant earlier. That violates stable reference. (ii) Identity drift: the later stage silently switches the target or identity condition while preserving the appearance of continuity. That again violates stable reference. (iii) Undoing the earlier constraint effect: the later stage erases or bypasses the earlier negative effect in a way that restores admissible use on the same scope. That violates the irreversible-commitment regime. These cases exhaust same-scope repair mechanisms. $\square$

Definition B.32 (Quotient interface for a same-side split). Let $(\text{Status}_D, \text{Good}_D)$ be a status interface and let $u \neq v$ be a same-side split. Write

$$q_{u,v} : V_D \rightarrow V_D / (u \sim v)$$

for the quotient map identifying u and v and leaving all other status values distinct. The corresponding quotient interface is

$$(\text{Status}_D^{u \sim v}, \text{Good}_D^{u \sim v}) := (q_{u,v} \circ \text{Status}_D, q_{u,v}(\text{Good}_D)).$$

The quotient changes the status presentation while retaining the original evaluated act and its original uses. Write $\rho : X_D^\bullet \rightarrow \mathcal{B}_D$ for the independently specified original act, context, scope, input,

and query-interpretation data, including retained lineage where required. This is the original base left unchanged below, not a new field containing an erased status or desired answer profile. The consumer interface is

$$J_{u,v}(a) = (\rho(a), q_{u,v}(\text{Status}_D(a))).$$

Let $b_D(a)$ be admission polarity and let $\mathcal{P}(a) = (p_\alpha(a))_{\alpha \in \mathcal{Q}_D}$ collect the complete original governing answers. The family is the original use family, including its applicability, warrants, continuation and history questions where required. Partial answers retain original definedness. A pending computation is not an established absence.

For these complete licensed uses, equal retained original identity and query data identify the same bearer and complete question. Each original reader supplies its individually warranted complete outcome. Determinacy therefore identifies their answers, including definedness, collectively for all original questions. Hence

$$\mathcal{P}(a) = F(\rho(a), b_D(a)).$$

This qualification is inherited, not imposed anew. Since u, v have the same polarity, a quotient-status polarity reader $\bar{b}$ exists, so

$$\mathcal{P}(a) = F(\rho(a), \bar{b}(q_{u,v}(\text{Status}_D(a)))).$$

All displayed equations involving Status_D are evaluated on its original domain of definition. The same-side quotient leaves that domain unchanged; its original definedness and absence questions remain among the retained original answers. Outside that original domain the quotient performs no change.

Every original answer thus factors through $J_{u,v}$. This is [2, Corollary 5.7 and Theorem 5.2], with [1, Theorems 6.2–6.3 and 8.7]. The complete profile need not factor through polarity alone.

Definition B.33 (Redescription-dependent, effective, and inert splits). Let $u \neq v$ be a same-side split for an admissibility-relevant status interface on a fixed domain D . In this trichotomy, quotienting means the joint retained-base interpretation above, and admissibility relevance concerns interpreted governing status. Here Status_D denotes the decoded governing status, not an arbitrary raw representation label. A raw serial number or realization label is kept in the separate realization record. Thus the comparison in (T1) concerns interpreted governing classifications; a raw-label difference alone does not trigger theorem B.35. Extra realization labels remain bookkeeping unless they purport to change an original governing answer.

- (T1) **Redescription-dependent.** The split is redescription-dependent if there exist acts $a, b \in X_D^\bullet$ with $a \sim b$ but $\text{Status}_D(a) \neq \text{Status}_D(b)$, with one of the two values equal to u and the other to v .
- (T2) **Effective.** Assuming the split is not redescription-dependent, it is effective if passing to the quotient interface changes some admissibility-relevant verdict on the fixed domain. Because u and v lie on the same side of the interface, present eligibility cannot change; so effectiveness is equivalently the claim that quotienting changes some later admissibility-relevant reuse or continuation verdict on D .
- (T3) **Inert.** Again assuming the split is not redescription-dependent, it is inert if passing to the quotient interface changes no admissibility-relevant verdict on the fixed domain.

Lemma B.34 (Exhaustiveness of the trichotomy). *Every same-side split is exactly one of the three kinds listed above.*

Proof. Take any same-side split $u \neq v$. Either the split changes under identity-preserving transformation or it does not. If it does, it is redescription-dependent. If it does not, then quotienting the two values either changes some admissibility-relevant verdict on the fixed domain or it does not. These are exactly the effective and inert cases. The three cases are mutually exclusive by construction. $\square$

Proposition B.35 (Redescription-dependent splits are impossible). *No admissibility-relevant same-side split may be redescription-dependent.*

Proof. If $a \sim b$ but $\text{Status}_D(a) \neq \text{Status}_D(b)$, then the same standing-bearing content receives different admissibility-relevant classifications merely by presentation. That contradicts reference stability and identity-preserving transformation invariance. $\square$

Lemma B.36 (Quotient-sensitivity of effective same-side splits). *Let $u \neq v$ be a same-side split for an admissibility-relevant status interface $(\text{Status}_D, \text{Good}_D)$ on a fixed domain D . Then the split is effective if and only if identifying u and v in the quotient interface changes at least one admissibility-relevant verdict on that same fixed domain. Equivalently, a same-side split is effective exactly when quotienting away the split changes some later admissibility-relevant reuse or continuation verdict while leaving the underlying acts, contexts, and all non- u/v status values unchanged.*

Proof. The forward direction is the definition of effectiveness. For the converse, if quotienting u and v together changes an admissibility-relevant verdict on the fixed domain, then passing from $(\text{Status}_D, \text{Good}_D)$ to $(\text{Status}_D^{u \sim v}, \text{Good}_D^{u \sim v})$ changes some admissibility-relevant verdict on that same domain. Again by definition, this is exactly effectiveness. The final sentence just unpacks the fact that u and v lie on the same side of the interface, so present eligibility is unchanged and any nontrivial change must occur in later admissibility-relevant reuse or continuation verdicts. $\square$

Lemma B.37 (Effective same-side splits perform hidden gate work). *Assume $u \neq v$ is a same-side split that is effective in the quotient-sensitive sense of theorem B.36. Then the changed verdict produced by passing to the quotient interface is attributable to the split itself rather than to ordinary act-level differences. Consequently:*

- (i) *if $u, v \in \text{Good}_D$, the split refines the admitted side in a way that changes later eligibility, hence constitutes deferred gate work or a second gate internal to the positive side; and*
- (ii) *if $u, v \in \text{Bad}_D$, the split marks at least one negative status as more promotion-susceptible than the other, hence opens a same-scope promotion channel.*

Proof. The underlying acts, contexts, and original question interpretations remain available through ρ . The inherited joint factorization preserves every original answer, including each original continuation's applicability and outcome. An effective split would require one of those answers to change while the same complete original use and its warrants remain fixed, a contradiction. Thus no admissible instance of the hypothesis exists: neither additional positive-side authority nor negative-side promotion is obtained. An implementation that loses an original required answer is not the faithful retained-base quotient under consideration. $\square$

Proposition B.38 (Effective same-side splits are impossible). *Under the AMetric boundary, the prior-gate theorem, and the no-promotion result above, no admissibility-relevant same-side split may be effective.*

Proof. A same-side merge retains polarity and the independently original base. The preceding factorization therefore retains every original governing answer. Effectiveness would change one of them, so no effective split exists in the complete licensed interface. No pre-boundary symmetry argument is applied to substantive distinctions retained in licensed post-boundary acts or contexts. $\square$

Proposition B.39 (Inert same-side splits collapse extensionally). *If a same-side split is inert, then quotienting its two values together changes no admissibility-relevant fact on the fixed domain. The split is therefore bookkeeping rather than governance.*

Proof. By definition, replacing $(\text{Status}_D, \text{Good}_D)$ by the quotient interface $(\text{Status}_D^{u \sim v}, \text{Good}_D^{u \sim v})$ changes no admissibility-relevant verdict on the fixed domain. Since u and v already lie on the same side of the interface, the quotient also preserves present eligibility. So identifying the two values changes notation only and performs no governance work. $\square$

Proposition B.40 (Binary collapse is derived from arbitrary status interfaces). *The bivalence argument does not assume binary status at the outset. It begins with an arbitrary admissibility-relevant status interface $(\text{Status}_D, \text{Good}_D)$ on a fixed domain and shows that any same-side refinement must be redescription-dependent, effective, or inert, hence impossible or extensionally eliminable.*

Proof. The arbitrariness of the starting point is built into the definition of admissibility-relevant status interface. The exhaustive trichotomy is theorem B.34, and its branches are handled by theorems B.35, B.38 and B.39. So the binary form is derived from arbitrary status interfaces rather than assumed in advance. $\square$

Theorem B.41 (Bivalence of the evaluated interface). *Let $(\text{Status}_D, \text{Good}_D)$ be any admissibility-relevant status interface on a fixed domain D . Then the interface is extensionally equivalent to a binary one. Equivalently, every admissibility-relevant status on D falls into exactly one of two classes:*

- (i) *admitted for commitment, reuse, or standing-bearing continuation on D ; or*
- (ii) *not admitted on D .*

Any additional same-side status distinctions are impossible or collapse to bookkeeping.

Proof. Assume the interface is not extensionally binary. Then there exists at least one admissibility-relevant same-side split $u \neq v$. By theorem B.34, the split is either redescription-dependent, effective, or inert. If it is redescription-dependent, theorem B.35 rules it out. If it is effective, theorem B.38 rules it out. If it is inert, theorem B.39 shows that it does no admissibility work and may be quotiented away. After collapsing every inert distinction, only two admissibility-relevant classes remain: eligible and ineligible. $\square$

Corollary B.42 (Unique gate classification). *Let $\text{Adm}_D, \text{Adm}'_D : X_D^\bullet \rightarrow \{0, 1\}$ be two binary admissibility predicates on the same fixed domain D . Assume both are lawful-redescription invariant and both are claimed to govern commitment and reuse on that same scope. Then they agree extensionally on $X_D^\bullet$.*

Proof. Assume, toward contradiction, that they disagree on some evaluated act a . Then one gate treats a as admitted while the other treats it as not admitted on the same fixed scope. If both verdicts

are allowed to stand, the same act is simultaneously admitted and not admitted on the same scope, contradicting theorem B.41. If a higher-order rule decides which gate applies, then admissibility depends on a selector over gates. That selector is an admissibility-relevant parameter, forbidden by theorems B.16 and B.17 unless one has declared an explicit scope change. So disagreement is impossible. $\square$

Definition B.43 (Reuse-stable admissibility). An evaluated act is *reuse-stably admissible* if it is admitted and every licensed same-scope continuation preserves the standing-relevant invariants of the target it is supposed to carry.

Proposition B.44 (Admitted same-scope continuation preserves the standing-relevant invariant bundle). *If an evaluated act is admitted on a fixed domain and later reused under a licensed same-scope continuation, then the continuation preserves the standing-relevant invariants of the target it is supposed to carry.*

Proof. Suppose admitted same-scope continuation fails to preserve the relevant invariant bundle. Because the continuation is scope-preserving, no explicit reset or domain change has been declared. So either the context silently retargets the act while presenting it as continuation of the same standing-bearing content, violating stable reference, or it consults some hidden selector or criterion that changes what counts as relevant midstream. The latter would reintroduce hidden gate work, which theorems B.16 and B.41 exclude. $\square$

Theorem B.45 (Standing emergence). *Let Stand_D be any predicate claimed to mark those evaluated acts on the fixed domain D that are eligible for standing-bearing continuation, commitment, or licensed reuse. Then for every $a \in X_D^\bullet$,*

$$\text{Stand}_D(a) = 1 \iff a \text{ is reuse-stably admissible on } D.$$

Equivalently, standing is not an extra interface-level classifier beyond admissibility on a fixed scope. It is admissibility as preserved through licensed same-scope continuation.

Proof. Suppose first that $\text{Stand}_D(a) = 1$. Because standing is claimed to govern eligibility for commitment and licensed reuse on the fixed scope, it is an admissibility-relevant status. By theorems B.41 and B.42, there is no additional admissibility-relevant classification on D except the unique binary gate. So $\text{Adm}_D(a) = 1$. Moreover, if a were not reuse-stable, some licensed same-scope continuation would alter the invariant bundle while still counting as continuation of the same standing-bearing content, contradicting theorem B.44. So every standing-positive act is reuse-stably admissible.

Conversely, assume that a is reuse-stably admissible. Then $\text{Adm}_D(a) = 1$ by definition. Suppose nevertheless that $\text{Stand}_D(a) = 0$. Since standing is supposed to govern commitment or later licensed continuation, this would split admitted acts into two substantive subclasses: those merely admitted and those admitted-with-standing. If the split affects what may later be done on the fixed scope, then it is an admissibility-relevant same-side split on the positive side, forbidden by theorem B.41. If the split affects nothing, then it is inert bookkeeping. So denying standing to a reuse-stably admissible act is impossible as a substantive classification. $\square$

Corollary B.46 (Standing is lawful-redescription invariant). *If $a \sim b$, then standing agrees on a and b .*

Proof. Lawful redescription preserves admissibility and the standing-relevant invariant bundle. So it preserves reuse-stable admissibility, and theorem B.45 gives the result. $\square$

Theorem B.47 (No same-act repair). *Let $a \in X_D^\bullet$. If a is not standing-positive on the fixed domain, then no internal same-scope post-processing that still treats the output as the same evaluated act can restore standing. Any legitimate improvement must instead proceed by explicit reset and fresh evaluation of a new act.*

Proof. Assume an internal same-scope step repairs a while still treating the result as the same evaluated act. If the repair changes admissibility, then one has a same-scope promotion from ineligible to eligible, impossible by theorem B.31. If the repair leaves admissibility unchanged but changes standing, then by theorem B.45 it changes reuse-stable admissibility. That can happen only by changing which invariant bundle is being preserved, which means silent retargeting, hidden reclassification of relevant invariants, or an undeclared reset. The first two contradict theorem B.44; the third is not same-act repair. $\square$

Theorem B.48 (No generators). *There is no internal operation T and no act $a \in X_D^\bullet$ with standing-failure such that $T(a)$ is standing-positive as a consequence of a alone and without fresh evaluation on D .*

Proof. If $T(a)$ is still treated as the same evaluated act as a , the claim is exactly same-act repair and is impossible by theorem B.47. If $T(a)$ is treated as a distinct act, then its positive standing depends on fresh evaluation of $T(a)$, not on the prior failed standing of a alone. So no generator exists. $\square$

Definition B.49 (Carrier attempt). *A carrier attempt on the fixed domain D is any derived predicate Q_D proposed to authorize standing independently of the unique admissibility gate. Formally, it is a rule of the form*

$$\text{Stand}_D(a) = 1 \text{ because } Q_D(a),$$

where Q_D is not merely an extensional renaming of reuse-stable admissibility.

Lemma B.50 (Any putatively independent authorizer induces gate work). *If a predicate Q_D is used as an independent authorizer to decide whether an evaluated act may enter standing-bearing continuation on D , then Q_D performs gate work. It either changes which acts are admitted for licensed reuse or refines the admitted side into a further substantive eligibility partition.*

Proof. If Q_D changes whether an act may enter standing-bearing continuation, then it directly changes eligibility. If instead all acts admitted by Adm_D remain admitted but only some of them are counted by Q_D as genuinely standing-positive, then Q_D introduces a substantive same-side split among admitted acts. No third possibility exists. $\square$

Theorem B.51 (No independent carriers). *No carrier attempt Q_D can serve as an independent authorizer of standing on the fixed domain D except by coinciding extensionally with reuse-stable admissibility. Any admissible use of Q_D must therefore be downstream bookkeeping inside an already gated interior.*

Proof. Let Q_D be a carrier attempt. By theorem B.50, if it is used as an independent authorizer then it performs gate work. If it changes who is admitted, it competes with the unique admissibility gate, contradicting theorem B.42. If it leaves admission unchanged but refines the admitted side substantively, it creates an admissibility-relevant same-side split on the positive side, contradicting theorem B.41. The only remaining possibility is extensional coincidence with reuse-stable admissibility, in which case Q_D is merely a renaming or bookkeeping layer. $\square$

Definition B.52 (Admissible interior). For a fixed domain D , the admissible interior is the class

$$\text{Int}_D := \{a \in X_D^\bullet : a \text{ is standing-positive on } D\}.$$

By theorem B.45, this is exactly the class of reuse-stably admissible acts on D .

Definition B.53 (Lawful equivalence of interiors). Two subsets $I_1, I_2 \subseteq X_D^\bullet$ are *equivalent under identity-preserving transformation* if they determine the same subset of the quotient $X_D^\bullet / \sim$. Equivalently, after identifying equivalent under identity-preserving transformation presentations, they contain exactly the same standing-positive acts.

An interior presentation here is a complete admissible interpretation of the original standing-membership questions on the original evaluated domain. Its decoded membership, not the contents of a selected display, enters the comparison. A selected view retaining complete original access is bookkeeping within that same interior. Extra realization data do not create another interior. Actual loss requires a required answer that cannot be recovered from the joint retained presentation. See [5, Theorem 3.1, Corollary 3.2, Theorem 8.1 and Section 8.1] and [4, Theorems 3.3, 5.1 and 5.5, and Corollaries 5.2 and 5.6].

Lemma B.54 (Difference of interiors yields a standing split). *Let $I_1, I_2 \subseteq X_D^\bullet$ be closed under identity-preserving transformation. If I_1 and I_2 are not equivalent under identity-preserving transformation, then there exists an evaluated act $a \in X_D^\bullet$ such that one interior treats a as standing-positive and the other does not.*

Proof. If the two interiors are not equivalent under identity-preserving transformation, then their images in the quotient $X_D^\bullet / \sim$ differ. So some lawful-redescription class belongs to one image but not the other. Choose any representative a of that class. Because both interiors are closed under identity-preserving transformation, membership of the class is equivalent to membership of a itself. $\square$

Theorem B.55 (Unique admissible interior). *For each fixed domain D , the admissible interior is unique up to lawful equivalence. More precisely, any two complete admissible presentations of that original interior give the same standing answer at every original evaluated act and therefore determine the same subset of $X_D^\bullet / \sim$.*

Proof. Fix an original evaluated act and its original standing-membership question. Each complete admissible presentation retains that question and its original warrant. Same-bearer, same-query conservation identifies their answers. Since the act was arbitrary, decoded membership agrees throughout the original domain, and therefore under lawful redescription. A selected display changes none of this when its complete original interpretation remains available. $\square$

Corollary B.56 (No plural admissible cores). *There is no genuine plurality of admissible interiors, either discrete or continuously parameterized.*

Proof. A discrete plurality gives two non-equivalent interiors and contradicts theorem B.55. A continuously parameterized plurality makes interior membership depend on an admissibility-relevant parameter, forbidden by theorems B.16 and B.17. $\square$

Theorem B.57 (Standing conservation and no same-domain portability without fresh gating). *For each fixed domain D , no internal operation of the licensed regime creates, amplifies, transfers, or recovers standing. Standing can only be preserved or destroyed.*

Proof. Recovery. Recovering standing for the same evaluated act is impossible by theorem B.47.

Creation. Creating standing from a failed act without fresh evaluation is impossible by theorem B.48.

Amplification. Amplification would require standing to admit degrees, but the governing interface is binary by theorem B.41, and theorem B.45 identifies standing with reuse-stable admissibility on that binary interface.

Transfer. Any internal transfer mechanism would need some predicate or object that carries standing from one act or context to another independently of fresh gating. That is exactly an attempt to introduce an independent carrier and is impossible by theorem B.51.

These four cases exhaust the standard ways of increasing standing. Therefore standing can only be preserved or destroyed. $\square$

Corollary B.58 (No cross-domain portability without fresh gating). *Standing in one domain does not by itself confer standing in another. Independent target gating on the target domain is required.*

Proof. Assume standing in one domain automatically confers standing in another without fresh target gating. Then either the source gate is being reused to evaluate acts on the target domain, which is illicit scope transport, or some derived property of “having standing in the source domain” is being used as an independent authorizer on the target domain, which is a carrier attempt. The first move is blocked by the same-domain scope discipline built into the fixed-domain setting; the second is blocked by theorem B.51. $\square$

Proposition B.59 (No semantic or infinitary standing import). *Truth, satisfaction, “all models,” reflection, and infinitary rule-power cannot serve as independent same-domain standing-bearing authorizers on the original fixed domain.*

Proof. If such a resource changes no standing or continuation fact, it is inert bookkeeping. If it does change them, then it acts as an independent same-domain authorizer and is therefore an inadmissible carrier by theorem B.51. If it can be made legitimate only after explicit scope enlargement, then it is a scope change rather than same-domain authority. These exhaust the possibilities. $\square$

Remark B.60 (Role of Appendix B). Appendix B is the place where the fixed domain is shown to have only one admissibility architecture. Once the AMetric boundary, no-deeper-invariant result, failure-class exhaustion, same-side split collapse, standing emergence, and conservation law are in place, no same-domain room remains for hidden positive-side refinement, repair channels, or admissibility-relevant parameterization.

C Standing quotient uniqueness and maximality

This appendix reconstructs the quotient layer used by the main Gödel result. It is formulated abstractly enough to apply to any fixed domain D .

Definition C.1 (Admissible histories and standing equivalence). Let $\mathcal{H}_D$ be the closure of primitive admissible tests on D under admissible same-scope continuation. For evaluated acts $a, b \in X_D^\bullet$, define

$$a \sim b \quad :\Longleftrightarrow \quad \text{Pred}(h; a) = \text{Pred}(h; b) \text{ for all } h \in \mathcal{H}_D.$$

This is the standing-equivalence relation of the fixed domain.

Definition C.2 (Standing presentation). A standing presentation for D is a triple (X, r, Pred_X) where:

- (i) X is a set of semantic standing states of D : their identity is their complete original standing meaning under $\mathcal{H}_D$, rather than the identity of a storage record, presentation label, or realization;
- (ii) $r : X_D^\bullet \rightarrow X$ is a surjection; and
- (iii) for every admissible history $h \in \mathcal{H}_D$ there is a prediction map $\text{Pred}_X(h; \cdot) : X \rightarrow Y_h$ such that

$$\text{Pred}(h; a) = \text{Pred}_X(h; r(a)) \quad (a \in X_D^\bullet).$$

The presentation is *sound* if $a \sim b \Rightarrow r(a) = r(b)$, and *complete* if $r(a) = r(b) \Rightarrow a \sim b$. The family $\mathcal{H}_D$ retains every original standing question for the complete use, including original applicability and absence meanings. This coverage is inherited, not obtained by adding unrelated contexts. A raw adequate representation $h_{\text{raw}} : X_D^\bullet \rightarrow R$ can retain additional realization information. Its semantic decoder $d : R \rightarrow X$ gives $r = d \circ h_{\text{raw}}$. Law-equivalent acts need not have equal raw records. Soundness concerns r , the semantic standing-state map. [4, Definition 3.2 and Theorem 3.3] separates complete adequacy from minimal semantic presentation.

Lemma C.3 (Completeness of standing presentations). *Every standing presentation is complete.*

Proof. If $r(a) = r(b)$, then for every admissible history h ,

$$\text{Pred}(h; a) = \text{Pred}_X(h; r(a)) = \text{Pred}_X(h; r(b)) = \text{Pred}(h; b).$$

So $a \sim b$ by definition. □

Lemma C.4 (Soundness forced by stable reference). *Assume the kernel of Appendix A. Then every standing presentation is sound.*

Proof. If $a \sim b$, all original history answers agree. These are the complete original standing questions defining the semantic states in X . Their meanings and individual warrants are inherited. Where presentations express the same original question differently, original same-bearer, same-query conservation identifies the warranted answers [1, Theorems 6.2–6.3]. Thus $r(a)$ and $r(b)$ realize the same complete original standing meaning and are the same semantic standing state. A raw record may still distinguish serial numbers, derivations, or other realization data. That difference remains in h_{raw} , not semantic r . Stable reference does not require erasing raw information. □

Lemma C.5 (Quotient universal property). *Let $q : X_D^\bullet \rightarrow X_D^\bullet / \sim$ be the quotient map. If $F : X_D^\bullet \rightarrow Y$ is constant on standing-equivalence classes, then there exists a unique map $\tilde{F} : X_D^\bullet / \sim \rightarrow Y$ with $F = \tilde{F} \circ q$.*

Proof. Define $\tilde{F}([a]) = F(a)$. This is well-defined because F is constant on equivalence classes. Uniqueness follows because q is surjective. $\square$

Theorem C.6 (Uniqueness of the standing quotient). *For the fixed domain D , the standing quotient $Q_D := X_D^\bullet / \sim$ is the unique same-domain standing state space compatible with admissible histories, up to unique isomorphism. More precisely, if (X, r, Pred_X) is any standing presentation, then there exists a unique bijection*

$$\phi : Q_D \rightarrow X, \quad \phi([a]) = r(a),$$

compatible with all admissible-history predictions.

Proof. By theorem C.4, r is constant on equivalence classes, so by theorem C.5 it factors uniquely through the quotient as $r = \phi \circ q$. The induced map is injective by theorem C.3: if $\phi([a]) = \phi([b])$, then $r(a) = r(b)$, hence $a \sim b$, so $[a] = [b]$. It is surjective because r is surjective. Compatibility with admissible-history predictions follows from the definition of standing presentation. $\square$

Definition C.7 (Same-domain standing-preserving map). A map $F : X_D^\bullet \rightarrow Y$ is *same-domain standing-preserving* if $a \sim b$ implies $F(a) = F(b)$ and if F changes neither standing classification nor admissible-history prediction on the fixed domain.

Theorem C.8 (Factorization through the standing quotient). *Every same-domain standing-preserving map factors uniquely through the standing quotient Q_D .*

Proof. If F is standing-preserving, then it is constant on standing-equivalence classes by definition. Apply theorem C.5. $\square$

Theorem C.9 (No proper same-domain standing-preserving extension). *There is no proper same-domain standing-preserving extension beyond the standing quotient. Equivalently: any allegedly richer same-domain structure either factors through the quotient and adds no new standing content, or else violates standing-equivalence and is not same-domain standing-preserving.*

Proof. Let F be an allegedly richer same-domain standing-preserving map. By theorem C.8, it factors uniquely through the quotient. Hence every standing-relevant distinction carried by F is already determined by the quotient classes. So F adds no new standing content beyond the quotient. If it did distinguish acts inside a quotient class, then it would not be constant on standing-equivalence classes and therefore would not be standing-preserving. $\square$

D Trajectory closure and irrecoverability

This appendix reconstructs the trajectory layer. It formalizes the sense in which closure-relevant distinctions must matter *in process*. The key dependency point is that legality is not introduced here as a second hidden governance act. It is the path-level realization of the already fixed same-domain licensing relation carried by the kernel-relative substrate.

The received retained-lineage substrate. The state and transition meanings are those of the original complete lineage use: a state retains its required predecessor and support record, and a legal edge is the original licensed continuation of that record. These are the standing and same-scope reachability of Recorded Substrate Theorem 2.22 in the trajectory source [6]. The receiving identification retains the original state, standing question, edge applicability, and warrants; it does not substitute a fresh sequence of separately evaluated objects. Accordingly, the source’s negative absorption is inherited with that complete use.

Kernel, Definition B.10, Theorem B.11, Lemma B.12, and Corollary B.13, gives an explicit retained-dependency realization [1]: required ancestors remain in the endpoint’s support closure. If $a(v)$ is a local condition, supported standing is $s(v) = \bigwedge_{u \in \text{Anc}^*(v)} a(u)$. Retained failed support therefore remains a failed conjunct at each dependent successor. This realizes the intended lineage meaning; it is not a claim that every arbitrary kernel-typed transition is a dependency edge. Structure, Theorem 7.2, and AMetric, Theorems 11.1–11.2 and Example 11.3, distinguish retained-prefix failure from negative absorption and give the exact endpoint criterion [2, 3].

Proposition D.1 (Legality is substrate-fixed, not a second governance act). *Within the fixed substrate, the licensed same-domain continuation relation is already typed by the kernel, the interface forcing layer, and the quotient closure results. Accordingly, a legal same-domain trajectory does not introduce a new standing-bearing classifier beneath coherence; it records only a finite path through already licensed same-domain continuations. Any purported complete admissibility-preserving replacement that changes the original legality answers on that same substrate would either refine standing beneath tensor content, introduce hidden gate work, depend on illicit redescription, or amount to a scope change.*

Proof. Original legal applicability and the corresponding joint warrants belong to the complete governing use. A faithful representation inherits those facts and therefore the same legal relation. This is the joint-use receipt of ATS, Theorem 7.5 and Corollary 7.6 [4]; statewise quotient equality alone is not substituted for it. A different raw record or a selected display can be bookkeeping while preserving access to that relation. A purported complete replacement that omits a required legal answer is nonfaithful; one that supplies a conflicting original answer cannot inherit its original warrant. Changing the governing conditions changes the comparison. Thus no independent legality decision is introduced beneath the received complete coherence use. $\square$

Definition D.2 (Legal same-domain trajectory). A legal same-domain trajectory is a finite nonempty sequence

$$\tau = (a_0, a_1, \dots, a_n)$$

of complete evaluated lineage states, with their original required support and predecessor records, such that each adjacent step is a licensed same-domain continuation already fixed by the substrate. Thus legality is typed prior to the coherence definition and is not available for later hidden refinement on the same substrate.

Definition D.3 (Coherent trajectory). A legal trajectory $\tau = (a_0, \dots, a_n)$ is *coherent* if every state on the trajectory is standing-positive.

Lemma D.4 (Prefix preservation). *Every prefix of a coherent trajectory is coherent.*

Proof. A prefix of a legal trajectory is legal by restriction to a shorter sequence. If every state on the full trajectory is standing-positive, then every state on any prefix is standing-positive as well. $\square$

Lemma D.5 (Failure witness characterization). *A legal trajectory is incoherent if and only if it contains at least one standing-failing state.*

Proof. Immediate from the definition of coherence. □

Theorem D.6 (No deferred coherence). *If a legal trajectory has an incoherent prefix, then every legal extension of that prefix within the same lineage is incoherent.*

Proof. An incoherent prefix contains a standing-failing state by theorem D.5. Every extension that retains the prefix retains that state as part of its lineage. Hence the extension also contains a failure witness and is incoherent. □

Theorem D.7 (Irrecoverability under admissible continuation). *If a legal same-domain trajectory contains a standing-failing state at stage i , then every later stage in that trajectory is standing-failing.*

Proof. The received state and relation are the original retained-lineage state and licensed continuation identified above. Recorded Substrate Theorem 2.22 of the trajectory source gives $S(a) = 0$ and $a \rightsquigarrow_D b$ implies $S(b) = 0$ [6]. Each suffix of the legal trajectory is such a same-scope continuation, so failure at stage i implies failure at every later stage. In the retained-dependency realization, this is exactly preservation of the failed required ancestor in each endpoint support. A fresh successful act outside that retained failed lineage does not change this conclusion. □

Theorem D.8 (Terminal-state criterion). *A legal trajectory is coherent if and only if its terminal state is standing-positive.*

Proof. If the trajectory is coherent, every state including the terminal one is standing-positive. Conversely, if the terminal state is standing-positive and some earlier state failed standing, then by theorem D.7 the terminal state would fail standing too. Therefore no earlier failure exists, and the trajectory is coherent. □

Remark D.9 (Why Appendix D matters to the Gödel bridge). Appendix D ensures that a closure-relevant distinction cannot be merely static or retrospective. To matter, it must alter what can be licensed in a same-domain constructional history. This is the trajectory side of the bridge used in theorem 3.7.

E Standing conservation as an identity-level invariant and the collapse of second equivalence relations

This appendix reconstructs the final equivalence layer. It proves that standing conservation is not an auxiliary rule but an invariant forced by the equivalence of admissibility and standing.

Theorem E.1 (Standing conservation as an identity-level invariant). *Once admissibility and standing are equivalent on the fixed domain, standing conservation is not an additional rule layered onto admissibility. It is an identity-level invariant of admissible classification itself. Equivalently, there exists no admissible determination that violates standing conservation, and no standing violation that preserves admissibility.*

Proof. Assume, for contradiction, that there exists an admissible determination that violates standing conservation. By Appendix B, standing on the fixed domain is exactly reuse-stable admissibility. So an admissible determination implies standing, while a violation of standing implies inadmissibility. The assumed determination is therefore both admissible and inadmissible, impossible. Conversely, if one assumes a violation of admissibility that preserves standing, then standing would hold without admissibility, contradicting theorem B.45. Hence any violation of one is a violation of the other. Conservation is therefore not an auxiliary overlay; it is the invariant form of admissibility under the standing-admissibility equivalence. $\square$

Closure-relevant equivalence here is semantic equivalence for the complete original standing use, as distinguished in Appendix C from equivalence on enriched raw records. Faithful bookkeeping may retain extra realization distinctions without a second semantic standing relation.

Theorem E.2 (No second closure-relevant equivalence relation). *There is no second closure-relevant equivalence relation on the fixed domain beside the standing-equivalence relation determined by admissible classification. Any alleged second closure-relevant equivalence either*

- (i) *collapses extensionally to the standing quotient; or*
- (ii) *violates the identity-level invariance of admissible classification and is therefore inadmissible.*

Proof. If the proposed semantic equivalence $\approx$ equals $\sim$, there is no second relation. Otherwise choose a pair in their symmetric difference. If $a \approx b$ but not $a \sim b$, an original admissible history distinguishes their answers, so identification loses an original required distinction. If $a \sim b$ but not $a \approx b$, all original history answers agree and theorem C.4 gives one semantic standing state. Separating the pair either retains only realization data, which is bookkeeping rather than a second semantic relation, or asserts an additional semantic distinction absent from the complete original use. This covers incomparable relations as well as strict refinements and coarsenings. No second complete semantic closure equivalence survives. $\square$

Remark E.3 (Role of Appendix E). Appendix E closes the final escape route for the main theorem. A hostile reader may try to say that the Gödel distinction lives in a different equivalence relation than the standing quotient. Theorem E.2 shows that no such second closure-relevant equivalence exists on the fixed domain.

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